Natural boundaries of anisotropic Ising susceptibility via real-cycle pole estimates

Dehao Lin
School of Physics, Sun Yat-sen University
ORCID: 0009-0001-4551-8490
lindh9@mail2.sysu.edu.cn

29 October 2026

Abstract

For each fixed positive coupling ratio, the high-temperature bulk susceptibility has a bounded lens as its continuation domain and the low-temperature connected pure-phase susceptibility has a right epigraph. We determine these domains from the endpoint equations and prove that their boundaries are natural. At resonant edges a fixed Vandermonde subproduct controls the differentiated particle tail. At negative high-temperature edges the odd momentum constraint displaces the first collision. A pole expansion on the real cycle and an angular concentration estimate give a beta period. At selected Taylor orders one particle index dominates the complete odd sum. For equal coupling, a phase-preserving tangential field gives tail bounds for both parities and both high-temperature boundary halves. The hyperbolic-sine covering transfers the isotropic obstructions to the physical temperature domains.

Introduction

The complex-temperature susceptibility of the square-lattice Ising model is governed by the interaction between singularities of its individual form-factor integrals and the convergence of their infinite sum. Orrick, Nickel, Guttmann and Perk [3, Section 3.2] analyzed the order dependence of the isotropic singularity amplitudes, while Boukraa, Hassani and Maillard [4, Section 6.2] studied the anisotropic cosine-grid singularity equations and their accumulation in two complex couplings. Fixing the positive physical coupling ratio selects a one-complex-variable temperature slice and its two physical continuations.

Endpoint profiles determine the physical domains and locate the regular arcs. The tail estimate on resonant arcs retains a fixed Vandermonde subproduct. For negative high-temperature arcs the proof compares pole integrals on the real cycle. Section 7 treats equal coupling directly by a phase-preserving contour argument for both parities.

AI Declaration

ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

Contents

1 Fixed-ratio slice and physical susceptibility 3
1.1 Field normalization and state . . . 3
1.2 Real-temperature spin form factors . . . 5
1.3 Symmetric double contours and every lattice site . . . 6
1.4 A holomorphic full sum on an independent spectral domain . . . 7

2 The two physical spectral components and their regular edges 8
2.1 Endpoint curves determine the physical components . . . 8
2.2 A single analytic profile difference controls every exception . . . 10
2.3 Canonical roots, prefactors, and the isotropic slice . . . 12

3 Regular spectral edges: whole amplitudes and the entire tail 12
3.1 A strict Schur bound, with the anisotropy retained . . . 13
3.2 All indices in the differentiated tail . . . 14
3.3 The first amplitude of the complete form factor . . . 15
3.4 All low-temperature signs; the high-temperature distinction . . . 16

4 Beta periods on the original real cycle at a negative HT edge 16
4.1 The exact pole and its direct Taylor estimate . . . 17
4.2 The complete residue density . . . 18
4.3 An angular estimate from the Gaussian period . . . 19
4.4 The exact complex quadratic period . . . 20
4.5 Uniform comparison and the two first windings . . . 21

5 Negative high-temperature edges: boundary jets and the complete odd sum 23
5.1 Setting and the existence of the boundary jets . . . 23
5.2 Analytic remainder estimates . . . 24
5.3 Uniform bounds for the winding sectors . . . 26
5.4 The exact first-winding model . . . 27
5.5 A discrete concavity estimate with controlled errors . . . 28
5.6 Selecting an integer order with one dominating index . . . 29
5.7 Completing the all-index comparison . . . 30

6 Unequal-coupling boundary theorem and isotropic specialization 31
6.1 Completing the unequal-coupling theorem . . . 31
6.2 Every fixed ratio and exceptional contact . . . 32
6.3 Isotropic coefficient identities . . . 32

7 The isotropic slice from its full physical product 33
7.1 Exact physical and sign identities . . . 34
7.2 Prime arithmetic and both HT geometries . . . 34
7.3 One analytic tail estimate for both isotropic parities . . . 35
7.3.1 A deformation of the complete constrained torus . . . 36
7.3.2 The mixed branch populations . . . 37
7.3.3 The regular inverse and a cell Gaussian bound . . . 38
7.3.4 The exact current and its joint jets . . . 40

7.3.5 Coarea with coupled occupancies 43
7.3.6 One sum for all higher particle orders 45
7.4 All lower coefficients and the complete finite complement 45
7.5 The first singular coefficient with its exact phase 47
7.6 Noncancellation for the infinite physical sums 49
7.7 The energy variable, the physical field, and covering 50

1 Fixed-ratio slice and physical susceptibility

Fix $J_x > 0$ and $\lambda = J_y/J_x \in (0, \infty)$. Throughout,

$$z = 2\beta J_x, \qquad S = \sinh z, \quad T = \sinh(\lambda z), \quad A = \cosh z \cosh(\lambda z), \quad p = ST. \tag{1.1}$$

The coupling ratio λ is held fixed in the complex continuation, during which T/S generally varies. There is a unique $z_c > 0$ with $\sinh z_c \sinh(\lambda z_c) = 1$. Define strictly increasing functions on $[0, \infty)$ by

$$P_\lambda(x) = \cosh x \tanh(\lambda x), \qquad Q_\lambda(x) = \cosh(\lambda x) \tanh x. \tag{1.2}$$

Their values at z_c are both one. Set

$$Y_\lambda = \frac{\pi}{2 \max(1, \lambda)},$$

$$H_\lambda(y) = \min\{P_\lambda^{-1}(\cos y), Q_\lambda^{-1}(\cos(\lambda y))\}, \qquad |y| < Y_\lambda, \tag{1.3}$$

$$L_\lambda(y) = \max\{P_\lambda^{-1}(|\cos y|), Q_\lambda^{-1}(|\cos(\lambda y)|)\}, \qquad y \in \mathbb{R}. \tag{1.4}$$

The domains of interest are specified independently of susceptibility:

$$\Omega_H = \{x + \mathrm{i}y : |y| < Y_\lambda, \ |x| < H_\lambda(y)\}, \qquad \Omega_L = \{x + \mathrm{i}y : x > L_\lambda(y)\}. \tag{1.5}$$

The value of H_λ at each endpoint is its continuous limit zero.

Theorem 1.1 (Full bulk susceptibility at every positive ratio)**.** *Let $0 < \lambda < \infty$. The high-temperature physical bulk susceptibility germ extends holomorphically to Ω_H, and $\partial\Omega_H$ is its natural boundary. The low-temperature physical pure-phase connected bulk susceptibility germ extends holomorphically to Ω_L, and the entire graph $\partial\Omega_L = \{L_\lambda(y) + \mathrm{i}y : y \in \mathbb{R}\}$ is its natural boundary. Here natural boundary means that no boundary point has a neighborhood carrying a holomorphic function agreeing with the indicated continuation on its domain side. Both statements concern the entire lattice and all particle indices. They apply separately to $\mathcal{X} = \beta^{-1}\chi$ and to χ with the field convention below.*

For unequal coupling, the local constants depend on the fixed ratio and the boundary point. At $\lambda = 1$, Section 7 proves the obstruction from the full physical product.

1.1 Field normalization and state

Let the magnetic field have energy units, so the finite-volume Hamiltonian contains $-h \sum_v \sigma_v$, with $\sigma_v \in \{-1, 1\}$ and magnetic moment set to one. Direct differentiation of the finite partition function gives

$$\left.\frac{\partial \langle \sigma_0 \rangle}{\partial h}\right|_{h=0} = \beta \sum_v \big(\langle \sigma_0 \sigma_v \rangle - \langle \sigma_0 \rangle \langle \sigma_v \rangle \big).$$

Thus the normalized response is the covariance sum $\mathcal{X}$, and $\chi = \beta\mathcal{X}$. A dimensionless field $\eta = \beta h$ instead has response $\mathcal{X}$. A specified nonunit magnetic moment multiplies both by its square.

On $0 < z < z_c$ the Gibbs state is unique and its magnetization is zero:

$$\mathcal{X}_H(z) = \sum_{(m,l)\in\mathbb{Z}^2} \langle \sigma_{00}\sigma_{ml}\rangle.$$

On $z > z_c$ take the thermodynamic plus state, or equivalently the minus state for this connected observable:

$$\mathcal{X}_L(z) = \sum_{(m,l)\in\mathbb{Z}^2} (\langle \sigma_{00}\sigma_{ml}\rangle_+ - M^2), \qquad M^2 = (1-p^{-2})^{1/4}.$$

The fourth root is positive on the displayed real interval. For the energy-field response in this phase, the derivative is the zero-field pure-phase linear response, equivalently the limit from $h > 0$ in the plus phase (or from $h < 0$ in the minus phase). The covariance formula is obtained with that phase fixed before taking the zero-field limit. A symmetric mixture has the same two-spin expectation but zero one-spin mean; using its zero mean as the subtraction would leave the nondecaying term M^2 and would define a different, divergent sum. This distinction is retained through the thermodynamic form-factor limit below.

The order facts needed here hold for arbitrary real site fields. Indeed the conditional probability of a plus spin at v is

$$\left(1 + \exp\left\{-2\beta\left(h_v + \sum_{u\sim v} J_{vu}\sigma_u\right)\right\}\right)^{-1},$$

with the fixed boundary spins included in the sum. It increases with every neighboring spin, boundary spin and field. Updating two ordered configurations at the same site with the same uniform random number therefore preserves their order. In a finite box, for fixed finite parameters, a full sweep has a uniformly positive probability of setting every spin to plus. Coupling two copies with those same parameters consequently proves convergence to the unique stationary law; the Gibbs law is stationary under each conditional update. Taking the limit in the ordered coupling proves finite-volume field and boundary monotonicity. Plus and minus volume limits inherit it and bound every Gibbs state, by conditioning such a state outside each finite box. Finite-support field changes of the plus state are exactly its bounded local tilts, as follows by taking the same limit in the finite-volume tilt formula. This proves the monotonicity statements also at negative fields, where GHS itself is not being applied. Differentiating finite-volume monotonicity in a single site field also gives nonnegative two-spin covariances; their volume limits remain nonnegative.

The covariance sum also equals the literal uniform-field derivative after the thermodynamic limit. Here is a precise justification once its finite absolute sum, proved below, is available. The GHS inequality for arbitrary nonnegative couplings and site fields [6, Theorem 1.5] says that truncated two-spin correlations decrease as any nonnegative site field increases. Pinning the spins removed from a larger plus box by fields tending to $+\infty$ consequently shows that the plus-box covariances increase with the box. Their sums converge monotonically to $\mathcal{X}$. If $m_V(h) = \langle\sigma_0\rangle_{V,+,h}$, GHS also makes m_V concave for $h \geq 0$. Thus

$$m_V(h) - m_V(0) \leq h m_V'(0) = \beta h \sum_{v\in V} \mathrm{Cov}_{V,+,0}(\sigma_0, \sigma_v) \leq \beta h \mathcal{X}.$$

Passing to the plus-state limit gives the upper right-derivative bound $m'(0+) \leq \beta\mathcal{X}$. Conversely, turn on h only in a finite set E in the infinite plus state. This is a bounded local change of measure, so its derivative at zero is exactly $\beta\sum_{v\in E}\mathrm{Cov}_{+,0}(\sigma_0,\sigma_v)$. Ferromagnetic field monotonicity makes

the uniform-field magnetization at least this local-field magnetization, with equality at $h = 0$. Letting E increase to the lattice gives the reverse derivative bound. Hence $m'(0+) = \beta\mathcal{X}$. In HT the zero-field plus and minus states coincide and have mean zero. Spin flip gives the same derivative for the minus state from $h < 0$. To check the plus state on that side as well, let $m_E(h)$ be the finite-support tilt of the zero-field state. Field monotonicity gives

$$m_-(h) \le m_+(h) \le m_E(h) \qquad (h < 0),$$

and division by the negative h reverses both inequalities. For each fixed E, the lower bound on the quotient tends to $\beta \sum_{v\in E} \mathrm{Cov}_0(\sigma_0, \sigma_v)$; the upper bound tends to $\beta\mathcal{X}$. Exhausting the lattice therefore proves the bilateral HT derivative. Other thermodynamic states lie between the extremal ones. In LT the stated right-sided plus-phase convention applies. The upper and lower derivative bounds identify the response after the thermodynamic limit; absolute summability controls the exhaustion of the lattice.

1.2 Real-temperature spin form factors

Away from criticality on either positive real interval set

$$B = T/S > 0, \qquad C = A/S, \qquad \cosh\gamma(q) = W(q) = C - B\cos q, \qquad \gamma(q) > 0.$$

Let $f_N(m, l)$ be

$$\frac{B^{N^2/2}}{N!} \int_{\mathbb{T}^N} \prod_{i=1}^{N} \frac{e^{-|m|\gamma(q_i)+\mathrm{i}lq_i}}{\sinh\gamma(q_i)} \prod_{i<j} \left[\frac{\sin((q_i - q_j)/2)}{\sinh((\gamma(q_i) + \gamma(q_j))/2)}\right]^2 \prod_{i=1}^{N} \frac{\mathrm{d}q_i}{2\pi}. \tag{1.6}$$

The infinite-plane correlations are

$$\langle\sigma_{00}\sigma_{ml}\rangle = \xi_H \sum_{N\ge1,\ N \text{ odd}} f_N(m, l), \qquad \xi_H = \frac{(1 - p^2)^{1/4}}{\sqrt{p}}, \tag{1.7}$$

$$\langle\sigma_{00}\sigma_{ml}\rangle_+ = \xi_L \left(1 + \sum_{N\ge2,\ N \text{ even}} f_N(m, l)\right), \qquad \xi_L = (1 - p^{-2})^{1/4} = M^2. \tag{1.8}$$

All powers are fixed by their positive real values. These are the anisotropic vacuum-to-state spin form factors of [1, 2], in the bulk normalization of [3], with the factor $B^{N^2/2}$ in the Nth term.

The finite-cylinder formulas in [1, (35)–(39)] concern arbitrary separations. The vacuum-to-state matrix elements are given in [2, Theorem 6.1] in the LT phase and in its Remark 6.2, equations (6.8)–(6.10), with the HT vacuum change. On a fixed real noncritical compact set, the one-body dispersion is analytic in a common angular strip and has a positive real gap. The finite-circumference correction functions η, Λ^{-1} and $\log\xi_T$ tend uniformly to zero: their sum-minus-integral expressions are controlled by the exponentially convergent trapezoidal rule for periodic analytic functions. They are uniformly bounded before the limit.

The particle limit is uniform as well. For real roots $0 < \rho(q) = e^{-\gamma(q)} \le r < 1$, the Schur factor $|(\rho_i - \rho_j)/(1 - \rho_i\rho_j)|$ is at most a constant strictly below one. The matched-Pfaffian argument proved in Section 3 applies to the discrete angular grids as an algebraic identity, with their normalized counting measures. Its single-body finite-size corrections cost only C^N. Consequently the finite-circumference coefficients, at each fixed separation, have a common bound $\exp(-cN^2 + O(N\log(N + 1)))$, including their factorial normalization. This is summable

independently of circumference, so the particle sum passes through the cylinder limit; taking each fixed-N Riemann-sum limit gives (1.6).

Every local cylinder limit is a translation-invariant, spin-flip invariant Gibbs state for these finite-range positive couplings. Apply [5, Theorem 1] with the vertex-transitive subgroup consisting of translations: unequal horizontal and vertical couplings satisfy all four stated hypotheses. Such a state is the equal mixture of the plus and minus states. Its even spin correlations are those of the plus state. This identifies the two-point function in (1.8); the connected pure-phase observable is obtained by removing the zero-particle term ξ_L.

The same expansion identifies the spontaneous magnetization. The maximal plus Gibbs state is extremal: in any convex decomposition into Gibbs states, each component is dominated by it; equality of the average forces equality on every increasing cylinder event, hence equality of the measures. Its tail is consequently trivial. Indeed, conditioning on a tail event with probability strictly between zero and one would decompose the state into two distinct Gibbs states. The reverse martingale theorem gives

$$\|\mathbb{E}_+(\sigma_0 \mid \sigma \text{ outside } V) - \langle\sigma_0\rangle_+\|_1 \longrightarrow 0.$$

For $v \notin V$ this norm bounds $|\operatorname{Cov}_+(\sigma_0, \sigma_v)|$, so the two-point function tends to $\langle\sigma_0\rangle_+^2$ at large separation. On a real noncritical interval, the positive dispersion gap and the uniform particle majorant just proved make every positive-particle term tend to zero as $|m| \to \infty$ with $l = 0$. Equations (1.7)–(1.8) therefore give $\langle\sigma_0\rangle_+ = 0$ in HT and $\langle\sigma_0\rangle_+^2 = \xi_L$ in LT. The plus mean is nonnegative by spin flip and extremal order. In HT the plus and minus means coincide at every site; their monotone coupling must then agree at every site almost surely. Thus the states coincide, and their extremal bounds prove uniqueness.

1.3 Symmetric double contours and every lattice site

For separate contour variables x, y put

$$D(x, y; z) = A - \frac{S}{2}(x + x^{-1}) - \frac{T}{2}(y + y^{-1}), \qquad \mathscr{A}(v, w) = \frac{v - w}{1 - vw}. \tag{1.9}$$

With $X = \prod_i x_i$ and $Y = \prod_i y_i$, the same correlation coefficient is

$$\frac{p^{N/2}}{N!} \oint \cdots \oint X^{|m|-1} Y^{|l|-1} \prod_{i<j} \mathscr{A}(x_i, x_j)\mathscr{A}(y_i, y_j) \prod_i \frac{\mathrm{d}x_i \, \mathrm{d}y_i}{(2\pi\mathrm{i})^2 D(x_i, y_i; z)}. \tag{1.10}$$

The circles are positively oriented and sufficiently close to, but inside, the unit circle. To check the normalization, fix $y = e^{\mathrm{i}q}$ and take the x residue at the interior root $\rho = e^{-\gamma}$. Since

$$D_x(\rho, y) = \frac{S(1 - \rho^2)}{2\rho^2}, \qquad \frac{\sqrt{p}}{D_x(\rho, y)} = \sqrt{B}\frac{\rho}{\sinh\gamma},$$

and subtraction of two dispersion equations gives

$$\mathscr{A}(\rho_i, \rho_j)\mathscr{A}(y_i, y_j) = -B\frac{(y_i - y_j)^2 \rho_i \rho_j}{y_i y_j (1 - \rho_i\rho_j)^2} = B\left[\frac{\sin((q_i - q_j)/2)}{\sinh((\gamma_i + \gamma_j)/2)}\right]^2 =: P_{ij}, \tag{1.11}$$

the residue factors supply $B^{N/2}$ and the pair factors supply $B^{N(N-1)/2}$. This recovers precisely $B^{N^2/2}$. The remaining factors in (1.10) give the stated Fourier and spatial decay factors. Each separation can be reflected independently at the fixed coupling ratio.

The exact full-site multiplier in the double contour is

$$\frac{(1+X)(1+Y)}{XY(1-X)(1-Y)} = \frac{1}{XY} + \frac{2(X^{-1}+Y^{-1})}{(1-X)(1-Y)}. \tag{1.12}$$

It is the product of the two sums $\sum_{m\in\mathbb{Z}} X^{|m|-1}$ and $\sum_{l\in\mathbb{Z}} Y^{|l|-1}$. It counts the origin once, each reflected axis correctly, and all four open quadrants. If C_N is the full-site coefficient and T_N the contour with the second numerator but without its factor two, then $C_N = f_N(0,0) + 2T_N$. The onsite sums are one in the HT phase and $1 - M^2$ after the LT subtraction.

Summing the Fourier variable and the geometric spatial series also gives the complete angular representation

$$C_N = \frac{1}{N!}\int_{\sum q_i = 0 \bmod 2\pi} G_N(q)\frac{1+Z}{1-Z}\prod_{i=1}^{N-1}\frac{\mathrm{d}q_i}{2\pi}, \quad G_N = B^{N/2}\prod_i \frac{1}{\sinh\gamma_i}\prod_{i<j} P_{ij}, \quad Z = \prod_i e^{-\gamma_i}. \tag{1.13}$$

There are exactly $N-1$ independent normalized angular measures, with $q_N = -\sum_{i<N} q_i$ modulo 2π.

1.4 A holomorphic full sum on an independent spectral domain

Define

$$\mathcal{S}_\lambda = \{z : \exists u, v \in [-1,1],\ A(z) = S(z)u + T(z)v\}. \tag{1.14}$$

It is exactly the set where (1.9) has a zero on the product of two unit circles. Section 2 proves that the components of $\mathbb{C}\setminus\mathcal{S}_\lambda$ containing $(0, z_c)$ and (z_c, ∞) are exactly (1.5), and that both are simply connected.

We record the scalar algebra used below. For even N the Schur identity is

$$\mathrm{Pf}\left(\frac{x_i - x_j}{1 - x_i x_j}\right)_{i,j=1}^{N} = \prod_{i<j}\frac{x_i - x_j}{1 - x_i x_j}.$$

For a short proof put $u_i = (1+x_i)/(1-x_i)$. Each entry becomes $(u_i - u_j)/(u_i + u_j)$. Induction in N compares the two rational functions at every possible simple pole $u_i + u_j = 0$: the residues agree by the induction hypothesis, because the two cross factors involving any third index multiply to one. Their difference has no poles, is homogeneous of degree zero, and is alternating, so it is zero. The case $N = 2$ starts the induction; removable special values follow by rational continuation. For odd N, append $x_{N+1} = -1$; all its entries are the removable constant one. This proves the matching identities at both parities.

Here is the needed convergence argument independently of any proposed natural-boundary conclusion. On a compact subset of the spectral complement the torus polynomial has a nonzero minimum modulus. Move both radii a small common amount inside one, within a product annulus where this gap persists. The resulting contours can be held fixed on a parameter neighborhood. Their Schur entries are bounded. The Schur Pfaffian identity, followed by Hadamard's determinant inequality, bounds the product of the two Pfaffians by $C^N N^{(N+1)/2}$, with the harmless extra odd dimension obtained by adjoining a constant ghost entry. After the $1/N!$ normalization and the bounded one-body and full-site factors this is summable in N. It is locally normally summable together with every fixed parameter derivative by Cauchy's inequality on a slightly larger neighborhood.

On a physical real compact set the same contours in (1.10) additionally give $r_x^{N|m|} r_y^{N|l|}$ with $r_x, r_y < 1$. This proves absolute summability in the lattice sites and particles and justifies their interchange. Thus (1.13) and the double contour define the same physical germs, with a proven full-site sum.

Write $F_N = p^{-N/2} C_N$ for the unscaled double contour, whose integrand uses (1.9) and (1.12). The complete sums are

$$\mathcal{X}_H = (1-p^2)^{1/4} \sum_{N\geq 1,\ N \text{ odd}} p^{(N-1)/2} F_N, \tag{1.15}$$

$$\mathcal{X}_L = (1-p^{-2})^{1/4} \sum_{N\geq 2,\ N \text{ even}} p^{N/2} F_N. \tag{1.16}$$

The powers inside both sums are integers. The identity $A^2 = (1+S^2)(1+T^2)$ implies that every $p^2 = 1$ point lies in $\mathcal{S}_\lambda$. Also, p never vanishes in $\Omega_L \subset \{\Re z > 0\}$; in Ω_H its only zero is the double zero at the origin. Hence the prefactors in these formulas have unique holomorphic branches on the specified simply connected domains, fixed by the real physical values. Formula (1.15) removes the apparent HT $1/\sqrt{p}$ singularity at zero; $\mathcal{X}_H(0) = 1$. In particular, the one-particle term is

$$\mathcal{X}_H^{(1)} = \frac{(1-p^2)^{1/4}}{A - S - T},$$

which gives $1 + 2\beta(J_x + J_y) + O(\beta^2)$ at the origin.

For the angular formulas the spectral root is globally specified by

$$\rho = W - \sqrt{W^2 - 1}, \qquad |\rho| < 1, \qquad W = (A - T\cos q)/S.$$

The square root has asymptotic W at infinity. For z in the spectral complement, W avoids $[-1, 1]$ for every real q, so the inside root is unique. At $S = 0$ its root equation $S\rho^2 - 2(A - T\cos q)\rho + S = 0$ gives its removable value zero. A local logarithm $\gamma = -\log \rho$ is used only where nonzero and needed. No unsupported global logarithm or change of transfer vacuum is imposed by periodic complex coupling shifts.

2 The two physical spectral components and their regular edges

Fix $\lambda > 0$. We continue to write $z = x + \mathrm{i}y$, $S = \sinh z$, $T = \sinh(\lambda z)$, $A = \cosh z \cosh(\lambda z)$, and $p = ST$. Define the spectral set by

$$\mathcal{S}_\lambda = \{z : A(z) = S(z)u + T(z)v \text{ for some } u, v \in [-1, 1]\}. \tag{2.1}$$

It is closed, by compactness of the square of representing pairs. It is the zero set of the double-contour polynomial with both contour variables on the unit circle. The four endpoint equations determine its two physical complement components.

2.1 Endpoint curves determine the physical components

Put

$$F_1(x, y) = \cosh^2(\lambda x)\cos^2 y - \cosh^2 x \sinh^2(\lambda x), \tag{2.2}$$

$$F_2(x, y) = \cosh^2 x \cos^2(\lambda y) - \sinh^2 x \cosh^2(\lambda x), \tag{2.3}$$

$$Z_\lambda = \{x + \mathrm{i}y : F_1(x, y) F_2(x, y) = 0\}. \tag{2.4}$$

Lemma 2.1 (Endpoint boundary criterion)**.** *For every positive fixed ratio,*

$$Z_\lambda \subset \mathcal{S}_\lambda, \qquad \partial\mathcal{S}_\lambda \subset Z_\lambda. \tag{2.5}$$

Proof. For $x > 0$, neither S nor $\tanh(z/2)$ vanishes or has a pole. For $\epsilon \in \{1, -1\}$ define

$$r_\epsilon(z) = e^{\epsilon\lambda z}\tanh(z/2), \qquad \frac{r_\epsilon + r_\epsilon^{-1}}{2} = \frac{A - \epsilon T}{S}. \tag{2.6}$$

The second identity follows by expanding the two exponentials. Moreover,

$$|\tanh(z/2)|^2 = \frac{\cosh x - \cos y}{\cosh x + \cos y}, \qquad |r_\epsilon| = 1 \iff \cos y = \epsilon\cosh x\tanh(\lambda x).$$

Thus an equation $A = Su + \epsilon T$ with $u \in [-1, 1]$ is equivalent to $F_1 = 0$, with the appropriate sign ϵ: the roots of $r + r^{-1} = 2u$ are on the unit circle, and conversely a unit root gives a real $u \in [-1, 1]$. Axis exchange gives the corresponding equivalence for $F_2 = 0$. At $x = 0$, S, T are imaginary and A is real, so membership in $\mathcal{S}_\lambda$ is exactly $\cos y\cos(\lambda y) = 0$; this is also $F_1F_2 = 0$. Reflection $z \mapsto -\overline{z}$ handles $x < 0$. In particular $Z_\lambda \subset \mathcal{S}_\lambda$ everywhere.

Now take $z_0 \in \partial\mathcal{S}_\lambda$. If the real-linear map $(u, v) \mapsto S(z_0)u + T(z_0)v$ has rank two and its representing pair is in the open square, the inverse real-linear map supplies representing pairs in that open square for all nearby z. This would make z_0 an interior point of $\mathcal{S}_\lambda$. Hence its representing pair has an endpoint coordinate. If the map has rank at most one, a nonempty affine fiber through the square meets the boundary of the square: move along a nonzero kernel direction until first reaching its boundary. Thus there is an endpoint representation in this case as well. Simultaneous zeros of S, T have $A = \pm 1$ and lie outside $\mathcal{S}_\lambda$. The endpoint equivalences place z_0 in Z_λ. □

Let $z_c > 0$ satisfy $\sinh z_c \sinh(\lambda z_c) = 1$. The functions

$$P_\lambda(x) = \cosh x\tanh(\lambda x), \qquad Q_\lambda(x) = \cosh(\lambda x)\tanh x$$

strictly increase from zero to infinity on $[0, \infty)$ and both equal one at z_c. In particular,

$$F_1 = \cosh^2(\lambda x)(\cos^2 y - P_\lambda(x)^2),$$
$$F_2 = \cosh^2 x(\cos^2(\lambda y) - Q_\lambda(x)^2).$$

Theorem 2.2 (Exact physical components)**.** *The components of $\mathbb{C} \setminus \mathcal{S}_\lambda$ containing $(0, z_c)$ and (z_c, ∞) are precisely the domains Ω_H and Ω_L in (1.5). They are simply connected. Their boundaries are the complete lens boundary and the complete graph $x = L_\lambda(y)$, respectively, and $0 \le L_\lambda \le z_c$.*

Proof. In the displayed lens, both F_1, F_2 are positive. In the displayed right epigraph they are both negative. Thus both domains are disjoint from Z_λ, whereas their entire boundaries lie in Z_λ. At the two lens tips this last statement follows from the first vanishing cosine and $x = 0$. The lens is star shaped about zero: its positive profile decreases with $|y|$. The epigraph is connected, since $L_\lambda \le z_c$.

By Lemma 2.1, $\partial\mathcal{S}_\lambda$ does not meet either connected domain. Therefore the intersection of $\mathcal{S}_\lambda$ with each domain is both relatively open and relatively closed. At zero it is absent, and it is also absent at every real $z > z_c$, because

$$(A - S - T)(A + S + T) = (ST - 1)^2 > 0$$

there, with all factors except possibly $A - S - T$ positive. Thus both domains lie in the spectral complement. Their entire boundaries lie in $\mathcal{S}_\lambda$, so no path in that complement can leave either domain. They are the claimed components.

Star shapedness gives simple connectivity of the lens. The homeomorphism $(x, y) \mapsto (x - L_\lambda(y), y)$ maps the epigraph to the right half-plane and gives its simple connectivity. □

2.2 A single analytic profile difference controls every exception

For this purpose extend P_λ, Q_λ to all real x. They are odd real-analytic bijections of $\mathbb{R}$, with everywhere positive derivatives. For example

$$P'_\lambda(x) = \sinh x \tanh(\lambda x) + \lambda \cosh x \operatorname{sech}^2(\lambda x) > 0.$$

Their inverse functions are therefore real analytic on all of $\mathbb{R}$. Set

$$R(y) = P_\lambda^{-1}(\cos y), \qquad U(y) = Q_\lambda^{-1}(\cos(\lambda y)), \qquad E(y) = R(y)^2 - U(y)^2. \tag{2.7}$$

In these terms $L_\lambda = \max(|R|, |U|)$, and $H_\lambda = \min(R, U)$ on the central strip. The intersections of the two endpoint graphs in $x \geq 0$ are exactly the zeros of E.

Lemma 2.3 (Isolated profile intersections)**.** *If $\lambda \neq 1$, the zeros of E are locally discrete on $\mathbb{R}$.*

Proof. Near zero the inverse-function Taylor expansions give

$$E(y) = z_c \left(\frac{\lambda^2}{Q'_\lambda(z_c)} - \frac{1}{P'_\lambda(z_c)} \right) y^2 + O(y^4) = z_c(\lambda \coth(\lambda z_c) - \coth z_c) y^2 + O(y^4).$$

For the second equality put $a = \sinh z_c$ and use $\sinh(\lambda z_c) = a^{-1}$: $P'_\lambda(z_c) = a(1 + \lambda a)/\cosh z_c$ and $Q'_\lambda(z_c) = (1+\lambda a)/(a \cosh z_c)$. The coefficient is nonzero for $\lambda \neq 1$, because $t \coth t$ strictly increases on $(0, \infty)$: its derivative has numerator $\sinh t \cosh t - t > 0$. Thus the globally real-analytic function E is not identically zero, so its zeros are locally discrete. □

The analytic-profile argument applies also at simultaneous cosine zeros. Such zeros exist exactly when $\lambda = r/s$ in reduced form with r, s odd, at $y = (2m + 1)\pi s/2$. They are precisely the points where the LT boundary can reach $x = 0$. At each such y_0, the two nonnegative endpoint profiles have leading terms $|y - y_0|/\lambda$ and $\lambda|y - y_0|$; the signed profiles in (2.7) remain analytic there.

We next check that every remaining physical boundary point has all the regularity properties used by the particle estimates. Write $D = \Im(\overline{S}T)$. Suppose $x > 0$, $F_1 = 0$ and $F_2 \neq 0$. By the endpoint criterion,

$$A = Su + \epsilon T, \qquad u \in [-1, 1], \qquad \epsilon = \operatorname{sign}(\cos y). \tag{2.8}$$

The following identity is decisive:

$$D^2(1 - u^2) = \tanh^2(\lambda x) F_2(x, y)^2. \tag{2.9}$$

Here is a short derivation including the endpoint cases. Put $K = \cosh z\, T - \epsilon \cosh(\lambda z)$. Direct expansion gives $K^2 = S^2(u^2 - 1)$, so

$$\left[\Re(\overline{K}T)\right]^2 = D^2(1 - u^2).$$

Writing $c = \cosh x$, $h = \sinh x$, $C = \cosh(\lambda x)$, $H = \sinh(\lambda x)$, the relation $F_1 = 0$ is $\cos y = \epsilon cH/C$, and hence

$$\begin{aligned}\Re(\overline{K}T) &= c\cos y\,|T|^2 - \epsilon HC \\ &= \epsilon\frac{H}{C}\left(c^2(C^2 - \cos^2(\lambda y)) - C^2\right) = -\epsilon\frac{H}{C}F_2.\end{aligned}$$

This proves (2.9). Its right side is positive, so $D \neq 0$ and $u \in (-1, 1)$.

The corresponding edge coordinate is also automatically nonramified. Indeed

$$\frac{r'_\epsilon}{r_\epsilon} = \epsilon\lambda + \frac{1}{S}, \qquad \Re\frac{r'_\epsilon}{r_\epsilon} = \epsilon\left(\lambda + \frac{\sinh x\sinh(\lambda x)\cosh(\lambda x)}{\cosh x}\right) \neq 0$$

on this endpoint curve. Thus $r'_\epsilon \neq 0$. Since $u = (r_\epsilon + r_\epsilon^{-1})/2 \in (-1, 1)$, also $r_\epsilon \neq \pm 1$, and

$$g_\epsilon(z) = \frac{A - \epsilon T}{S} \quad \text{has} \quad g'_\epsilon(z) \neq 0.$$

Axis exchange gives the same statements when $F_2 = 0$, $F_1 \neq 0$; reflection gives them for $x < 0$.

Proposition 2.4 (Dense regular physical boundary arcs)**.** *For every $\lambda \neq 1$, the regular edge points used in the particle arguments are dense in both physical boundaries. Their excluded set is locally discrete. At each such regular point the two spectral directions are real-linearly independent, exactly one representing cosine is ± 1, the other is strictly inside $(-1, 1)$, and its holomorphic edge coordinate is nonramified.*

Proof. Away from the isolated profile intersections and the two imaginary lens tips, the active boundary graph has $x \neq 0$ and exactly one F_i equal to zero. The preceding calculation gives all the stated properties. The active profile is analytic there; the boundary has no isolated point because it is a continuous graph or a lens curve. Removing a locally discrete subset therefore leaves a dense set. Finally, at a rank-two endpoint representation, the Cramer coordinates vary continuously and the strict interior coordinate stays interior. More explicitly, on an F_1 chart the real representing coordinate $v(z)$ satisfies $v(z) - \epsilon = |S(z)|^2\Im g_\epsilon(z)/D(z)$. Thus the equation that this coordinate equals its endpoint is the nondegenerate real analytic arc just described; the spectral set occupies one closed side and the complement occupies the other. This verifies the local domain-side condition used below. □

On the right side of the HT lens, both cosines are positive, so the endpoint sign is positive. Reflection $z \mapsto -\overline{z}$ sends (u, v) to $(-u, -v)$ and gives negative endpoints on its left side. On the LT boundary the sign is $\operatorname{sign}(\cos y)$ for an F_1 edge, and $\operatorname{sign}(\cos(\lambda y))$ for an F_2 edge. Thus all the signs treated below occur on the specified physical components.

Each nonramified edge coordinate maps a sufficiently small regular arc onto a real interval in $(-1, 1)$. Values $\cos(2\pi k/q)$ with q an odd prime and $1 \leq k \leq (q-1)/2$ are dense in that interval, since the mesh tends to zero along the primes. At a positive endpoint the first odd resonance is q and the first even resonance is $2q$. At a negative endpoint all-endpoint momentum conservation requires an even particle order, whose first resonance is again $2q$. The odd sum at negative edges is treated by the boundary-jet argument.

2.3 Canonical roots, prefactors, and the isotropic slice

Lemma 2.5 (Canonical spectral root)**.** *For $z \notin \mathcal{S}_\lambda$ and real q, the equation*

$$S\rho^2 - 2(A - T\cos q)\rho + S = 0$$

has a unique root with $|\rho| < 1$, interpreted as $\rho = 0$ if $S = 0$. The root is holomorphic in z, locally jointly holomorphic with the angle, and has a uniform modulus gap below one on each compact parameter set and the real angle circle. On both positive noncritical real intervals it is the physical positive decaying root.

Proof. For $S \neq 0$, put $W = (A - T\cos q)/S$. The spectral-set definition gives $W \notin [-1, 1]$. The reciprocal roots of $\rho + \rho^{-1} = 2W$ cannot be on the unit circle, so exactly one is inside. It is simple, and the implicit-function theorem makes it locally holomorphic. Uniqueness glues these local roots. At $S = 0$, the coefficient $A - T\cos q$ is nonzero, so the same theorem extends the root through zero. Compactness gives the uniform gap and a thin complex angular neighborhood. Finally $A - S - T > 0$ on the positive noncritical real line by the identity used above, so $W(q) > 1$ and the interior root is positive. □

Where ρ is nonzero, write $\gamma = -\log\rho$ locally. The identity

$$A^2 - (\epsilon_1 S + \epsilon_2 T)^2 = (p - \epsilon_1\epsilon_2)^2$$

shows $\{p^2 = 1\} \subset \mathcal{S}_\lambda$. The only zero of either S or T in the HT lens is the common zero at $z = 0$; neither vanishes in the LT epigraph, where $x > 0$. Thus the physical phase prefactors have the holomorphic branches asserted in Section 1; the integer-power HT formula handles the origin.

At $\lambda = 1$, write $s = \sinh z$. For $s \neq 0$,

$$A/S = s + s^{-1} = u + v \in [-2, 2] \quad \Longleftrightarrow \quad |s| = 1,$$

so

$$\mathcal{S}_1 = \{z : |\sinh z| = 1\}, \qquad H_1(y) = \operatorname{arsinh}(\cos y), \quad |y| < \pi/2, \qquad L_1(y) = \operatorname{arsinh}(|\cos y|). \tag{2.10}$$

Section 7 gives the biholomorphism on the HT lens, the universal covering on the LT epigraph, and the transfer of boundary singularities through these maps.

3 Regular spectral edges: whole amplitudes and the entire tail

At a regular boundary point exactly one of the two real spectral coordinates is $+1$ or -1, the other lies strictly between them, and the corresponding edge map is a nonramified holomorphic local coordinate. Interchange the transfer axes when necessary. First take the endpoint $q_0 = 0$ and write

$$W(q; u) = b + u + B(u)(1 - \cos q), \qquad b = \cos\phi, \quad 0 < \phi < \pi, \quad \Im B(0) > 0. \tag{3.1}$$

Here $u = W(0) - b$ vanishes at the selected point. The opposite sign of $\Im B$ is obtained by complex conjugation. On the edge, $W(q)$ is in the upper half-plane except at $q = 0$. Its canonical root $\rho(q) = e^{-\gamma(q)}$ lies in the lower half-disk, with $\rho(0) = e^{-\mathrm{i}\phi}$. Compactness gives

$$\Im\rho(q) \leq -\delta < 0, \qquad |\rho(q)| \leq 1 \quad (q \in \mathbb{R}). \tag{3.2}$$

There is no real-angle point with $W = \pm 1$. The root, a local γ, and all one-body and canceled-pair denominators are therefore holomorphic on a common fixed complex parameter and angular neighborhood. On the physical-domain approach $u = \mathrm{i}\varepsilon$, $\varepsilon > 0$,

$$\Re\gamma(q; \mathrm{i}\varepsilon) \geq c\big(\varepsilon + \operatorname{dist}(q, 0)^2\big). \tag{3.3}$$

Indeed $\Im W = \varepsilon + \Im B(\mathrm{i}\varepsilon)(1 - \cos q)$ is bounded below by that expression, while $\Im W = \sinh(\Re\gamma)\sin(\Im\gamma)$ on the compact chosen branch. The same local side belongs to the physical component bordering the regular edge, by the geometry of Section 2.

3.1 A strict Schur bound, with the anisotropy retained

The elementary identity

$$|1 - vw|^2 - |v - w|^2 = (1 - |v|^2)(1 - |w|^2) + 4\Im v \Im w \tag{3.4}$$

and (3.2) give a constant $\rho_0 < 1$ such that $|\mathscr{A}(\rho_i, \rho_j)| \leq \rho_0$ for every pair of real angles. By continuity a slightly larger constant still below one works on a fixed complex parameter neighborhood. This assertion continues to hold there even where a separate inequality $|\rho_i| \leq 1$ no longer does. Every complete canceled pair P_{ij} is analytic and bounded on the same neighborhood.

The complete canceled pairs can exceed one. For example, on (3.1),

$$P(q, -q) = \frac{B \sin^2 q}{[b + B(1 - \cos q)]^2 - 1}.$$

When $\Im B$ is small and the real part of the bracket is close to one, this can be arbitrarily large. The proof below retains the strict bound only for the uncanceled ρ-Schur factors.

Lemma 3.1 (Matched-Pfaffian numerator estimate)**.** *For every fixed integer $r \geq 0$, on a common parameter disk,*

$$|\partial_u^r G_N(q; u)| \leq C_r \exp\{-\kappa N^2 + CN \log(N + 1)\}$$

uniformly in the real angles and all positive N. The constants may depend on the edge but not on N.

Proof. For even N, expand only the Schur Pfaffian in the $y_i = e^{\mathrm{i}q_i}$ variables. Each of its $(N-1)!!$ matchings contains $N/2$ edges. Combine each matched y factor with its corresponding ρ factor using (1.11); bound that complete pair by a constant C. The other $N(N-2)/2$ factors are bounded by ρ_0. Thus

$$\left| \prod_{i<j} P_{ij} \right| \leq (N-1)!!\, C^{N/2} \rho_0^{N(N-2)/2}.$$

For odd N, adjoin a ghost variable -1 to the y Pfaffian. Its Schur entry with every real variable is the removable constant one. There are $N!!$ matchings, with $(N-1)/2$ real matched edges and one ghost edge, leaving $(N-1)^2/2$ uncanceled ρ factors. This gives

$$\left| \prod_{i<j} P_{ij} \right| \leq N!!\, C^{(N-1)/2} \rho_0^{(N-1)^2/2}.$$

The identities initially hold away from reciprocal y pairs and extend by the analytic canceled formula. No division by a vanishing ρ factor occurs. One-body factors, including $B^{N/2}$, cost C^N. These are Gaussian bounds in N up to $\exp(O(N \log(N+1)))$. Cauchy's inequality on the common disk gives the jets of G_N. The product kernel $(1 + Z)/(1 - Z)$ is estimated separately below. □

3.2 All indices in the differentiated tail

Put $Q_0(q) = \sum_i \operatorname{dist}(q_i, 0)^2$. Equation (3.3) implies

$$|1 - Z| \geq 1 - |Z| \geq c \min\{1, N\varepsilon + Q_0(q)\}.$$

All fixed parameter derivatives of γ are bounded. Differentiating its exponential and the rational factor gives, for every fixed j and $0 \leq r \leq j$,

$$\left|\partial_u^r \frac{1+Z}{1-Z}\right| \leq C_j N^j \big(1 + Q_0(q)^{-j-1}\big). \tag{3.5}$$

We retain a fixed number of numerator collision factors before using Lemma 3.1. This gives one integrable majorant for every higher particle index.

Fix an integer $m \geq 1$, and write

$$\mathcal{V}_m(q) = \prod_{1 \leq i < \ell \leq m} \sin^2((q_i - q_\ell)/2).$$

For $N \geq m$, the quotient $G_N/\mathcal{V}_m$ has an analytic extension through the zeros of $\mathcal{V}_m$. Indeed, each complete pair internal to the first m labels is replaced by $B/\sinh^2((\gamma_i + \gamma_\ell)/2)$. These pairs and the first m one-body factors are bounded by a constant depending only on m. The $m(N-m)$ cross pairs cost $C^{m(N-m)}$, and the complementary numerator on the last $N-m$ labels satisfies Lemma 3.1; the empty numerator is interpreted as one. Thus, on a common complex parameter disk,

$$|G_N/\mathcal{V}_m| \leq C_m \exp\{-\kappa(N-m)^2 + C(N-m)\log(N-m+1) + Cm(N-m)\}.$$

Cauchy's estimate on a fixed smaller disk, applied to this analytic quotient, proves for every fixed $r \geq 0$ that

$$|\partial_u^r G_N(q;u)| \leq C_{m,r} \mathcal{V}_m(q) \exp\{-\kappa N^2 + C_m N \log(N+1)\}. \tag{3.6}$$

All constants are independent of N and of the real angles.

Let n be the first resonant index of the selected parity, and set

$$k = \frac{n^2-1}{2} \quad (n \text{ odd}), \qquad k = \frac{n^2}{2} - 1 \quad (n \text{ even}), \qquad m = n+1.$$

Every larger same-parity N satisfies $N \geq n+2 = m+1$, so the first m angles are independent integration variables. Choose their representatives in $[-\pi, \pi]$ and put $R^2 = \sum_{i=1}^m q_i^2$. Then $Q_0 \geq R^2$ and $\mathcal{V}_m \leq C_m R^{m(m-1)}$. The fixed m-dimensional majorant

$$\mathcal{V}_m(q)\big(1 + R^{-2j-2}\big)$$

is integrable for every $0 \leq j \leq k$: near zero its radial power is $R^{m^2-2j-3}\,\mathrm{d}R$, and $m^2 > 2j+2$. Every unused normalized angular measure has mass one. Applying the product rule to (1.13), and using (3.5) and (3.6), therefore gives

$$|\partial_u^j C_N(\mathrm{i}\varepsilon)| \leq \frac{C_{n,j} N^j}{N!} \exp\{-\kappa N^2 + C_n N \log(N+1)\}, \qquad N > n, \quad N \equiv n \pmod 2, \quad 0 \leq j \leq k. \tag{3.7}$$

The bound is summable in N, uniformly as $\varepsilon \downarrow 0$. Hence

$$\sum_{\substack{N > n \\ N \text{ of the selected parity}}} |\partial_u^j C_N(\mathrm{i}\varepsilon)| = O_j(1), \qquad 0 \leq j \leq k. \tag{3.8}$$

The same summable majorant controls every higher particle order of the selected parity and every derivative through order k.

3.3 The first amplitude of the complete form factor

Choose $\phi = 2\pi a/p_0$ with p_0 an odd prime and $1 \le a \le (p_0-1)/2$. Then the first odd resonance is $n = p_0$ and the first even resonance is $n = 2p_0$. For all lower indices of the selected parity, the only possible unit-modulus product configuration has $Z = e^{-\mathrm{i}N\phi} \neq 1$, so those entire form factors are holomorphic on a neighborhood of the selected point.

The Gaussian Vandermonde period in independent-angle coordinates is

$$G_n^{\mathrm{Gauss}} := \int_{q_n=-\sum_{i<n} q_i} e^{-Q(q)}\Delta(q)^2\,\mathrm{d}q_1\cdots\mathrm{d}q_{n-1} = \frac{n!\,\pi^{(n-1)/2}P_n}{\sqrt{n}}, \qquad P_n = \prod_{r=1}^{n-1}\frac{r!}{2^r}. \tag{3.9}$$

Here is an elementary evaluation of the Gaussian period. The monic polynomials

$$H_r(x) = (-1)^r 2^{-r} e^{x^2}\frac{\mathrm{d}^r}{\mathrm{d}x^r}e^{-x^2}$$

are orthogonal for $e^{-x^2}\,\mathrm{d}x$: integrating by parts r times against a polynomial of degree less than r gives zero. The same calculation against H_r gives $\int H_r(x)^2 e^{-x^2}\,\mathrm{d}x = 2^{-r}r!\sqrt{\pi}$. Since $\left(\det[H_{r-1}(x_i)]_{r,i=1}^n\right)^2 = \Delta(x)^2$, expansion of the two determinants and orthogonality give the ordinary n-variable integral $n!\pi^{n/2}P_n$. Put $x_i = q_i + m$ with $\sum q_i = 0$. The coordinate Jacobian is n, while integration in m gives $\sqrt{\pi/n}$. Dividing by their product proves (3.9). Euclidean surface measure on the zero-sum hyperplane is $\sqrt{n}$ times this coordinate measure.

Write $s_\phi = \sin\phi$ and $\alpha = (n^2-1)/2$. The singular part of the complete first form factor, on its unique collision chart, has leading expression

$$\frac{2\mathrm{i}^{1-n^2}B(0)^{n^2/2}}{n!(2\pi)^{n-1}2^{n(n-1)}s_\phi^{n^2-1}}\int\frac{\Delta(q)^2}{nu+(B(0)/2)Q(q)}\,\mathrm{d}q_1\cdots\mathrm{d}q_{n-1}. \tag{3.10}$$

This follows directly from

$$\gamma(q;u) = \mathrm{i}\phi - \frac{\mathrm{i}}{s_\phi}\left(u + \frac{B(0)}{2}q^2\right) + O(u^2+uq^2+q^4)$$

and the sine divided differences in the numerator. The collision chart is unique, and its complement in the fixed-n torus integral is analytic.

Differentiate (3.10) to the specified order and scale $q = \sqrt{\varepsilon}\,v$ on $u = \mathrm{i}\varepsilon$. The coefficient of Q in γ has positive real part, so the Gaussian/Laplace period is well-defined with its prescribed branch. The radial tails after the top derivative have radial integrands $O(R^{-3})$ for odd n and $O(R^{-2})$ for even n; their integrated tails are respectively $O(R^{-2})$ and $O(R^{-1})$. This gives dominated convergence. The error estimate can also be made directly before scaling. After factoring out $\Delta(q)^2$, the analytic numerator differs from its center value by $O(|u|+Q)$, and the pole denominator differs from $nu + B(0)Q/2$ by $O((|u|+Q)^2)$. On $u = \mathrm{i}\varepsilon$ both denominators have modulus at least $c(\varepsilon+Q)$ in a sufficiently small fixed chart. Differentiating the exact difference of the two quotients k times therefore bounds its absolute value by $C\Delta(q)^2(\varepsilon+Q)^{-k}$. Its radial integral is $O(1+|\log\varepsilon|)$ for odd n and $O(1)$ for even n. Multiplication by ε or $\sqrt{\varepsilon}$, respectively, sends this error to zero. The off-chart terms are holomorphic on a fixed neighborhood. This proves the differentiated asymptotic without an unbounded formal expansion. Using (3.9) and the Laplace

integral yields

$$\lim_{u\to 0,\ u=\mathrm{i}\varepsilon} u\,\partial_u^k C_n(u) = \frac{2^{1-k}\sqrt{B(0)}P_n n^{k-3/2}}{\pi^{(n-1)/2}\sin^{2k}\phi} \neq 0, \qquad k=\frac{n^2-1}{2},\quad n \text{ odd}, \tag{3.11}$$

$$\lim_{u\to 0,\ u=\mathrm{i}\varepsilon} u^{1/2}\partial_u^k C_n(u) = -\mathrm{i}\frac{2^{1-\alpha}\sqrt{B(0)}P_n n^{k-1}}{\pi^{(n-2)/2}\sin^{n^2-1}\phi} \neq 0, \qquad k=\frac{n^2}{2}-1,\quad n \text{ even}. \tag{3.12}$$

Here $u^{1/2}=e^{\mathrm{i}\pi/4}\sqrt{\varepsilon}$, and $\sqrt{B(0)}$ is fixed by the beta-period continuation from $\Re(-\mathrm{i}B(0)/(2s_\phi))>0$. For even n this sign is determined by the period with the original integer power of B.

For the odd physical HT coefficients the square-root convention is particularly explicit. On Ω_H, both $\cos y$ and $\cos(\lambda y)$ are positive, and $\sin y$ and $\sin(\lambda y)$ have the same sign. Consequently

$$\Re(T\overline{S}) = \sinh x\sinh(\lambda x)\cos y\cos(\lambda y)+\cosh x\cosh(\lambda x)\sin y\sin(\lambda y)>0$$

away from the origin, where B has its positive removable value λ. Thus $\Re B>0$ throughout the HT lens; its physical square root is the principal branch. The same is true after axis exchange, since $\Re(1/B)>0$. This fixes the odd amplitudes on both transfer-axis charts without an additional sign assumption.

The lower derivatives of the even first term are bounded. For the odd first term, the $(k-1)$st derivative can be logarithmic, and all lower derivatives are still $o(u^{-1})$. Together with (3.8) this gives the leading singularity of the full particle series. The physical prefactor and $\beta=z/(2J_x)$ are nonzero at a regular selected point, and the product-rule contributions from lower derivatives are smaller. A nonramified local change from u to z also preserves the nonzero top coefficient.

3.4 All low-temperature signs; the high-temperature distinction

At a negative endpoint put $q=\pi+q'$. Then $W=b+u+(-B)(1-\cos q')$. For even N, the constraint $\sum q_i=0 \pmod{2\pi}$ remains exactly the same constraint for q_i', and $B^{N^2/2}=(-B)^{N^2/2}$ because $N^2/2$ is even. Thus the whole even argument, including its tail and amplitude, applies to both endpoint signs. The other pair of edges is handled by exchanging the axes in the symmetric double contour. Rational phase values of the form above are dense on every regular arc. Section 2 then supplies the density needed on all of $\partial\Omega_L$.

For odd N, the negative endpoint has a different constraint: $\sum_i(q_i-\pi)$ is an odd multiple of π. In particular, with representatives $\delta_i=q_i-\pi\in[-\pi,\pi]$,

$$\sum_i \delta_i^2 \geq \frac{\pi^2}{N}. \tag{3.13}$$

There is no exact odd all-endpoint collision there. The same Gaussian numerator bound and (3.13) imply that the sum of every fixed parameter derivative is bounded as the negative edge is approached. This proves existence of all boundary jets, but gives no positive radius for their Taylor series. The following two sections resolve this separate infinite-index issue by proving zero radius at a dense rational-phase family.

4 Beta periods on the original real cycle at a negative HT edge

At a negative endpoint the odd momentum constraint displaces the closest collision by π/n. On the real integration cycle, the Vandermonde angular measure controls the cubic and quartic errors in the quadratic pole approximation. The angular estimate uses the constrained Gaussian period, gamma moments, and the arithmetic–geometric mean inequality.

4.1 The exact pole and its direct Taylor estimate

Fix a regular negative endpoint and exchange the transfer axes if needed. Use the nonramified coordinate

$$t = \frac{C + B - b}{B}, \qquad C = A/S, \qquad W(\pi + \delta; t) = b + B(t)(t - 1 + \cos\delta), \qquad b = \cos\phi. \tag{4.1}$$

After conjugation if necessary, assume

$$0 < \phi < \pi, \qquad \Im B(0) < 0, \qquad \phi = 2\pi r/p, \qquad (r, p) = 1, \quad p \geq 3 \text{ odd}. \tag{4.2}$$

The function B is holomorphic and nonzero on a fixed disk. Set $s = \sin\phi$ and choose the holomorphic branch

$$g(t, \delta) = \operatorname{arcosh} W(\pi + \delta; t), \qquad g(0, 0) = \mathrm{i}\phi.$$

It is even in δ, and g_t is bounded away from zero on a sufficiently small fixed bidisk. Constants below may depend on these fixed local data, but not on the particle number or Taylor order.

Let n tend to infinity through odd multiples of p, and put

$$d_n = \frac{\pi}{n}, \qquad \tau_n = 1 - \cos d_n, \qquad B_n = B(\tau_n), \qquad \alpha_n = \frac{n^2 - 1}{2}, \tag{4.3}$$

$$h_n = \cos d_n - \frac{B_n b}{s^2} \sin^2 d_n = 1 + O(n^{-2}), \qquad a_n = \Re h_n > 0. \tag{4.4}$$

In the first winding write $\delta_i = d_n + v_i$, with

$$V_n = \{v \in \mathbb{C}^n : \sum_i v_i = 0\}, \qquad Q(v) = \sum_i v_i^2.$$

The real integration measure is $\mathrm{d}v_1 \cdots \mathrm{d}v_{n-1}$ with $v_n = -\sum_{i<n} v_i$. When v is real, write $Q = Q(v)$ and $\Delta(v) = \prod_{i<k}(v_i - v_k)$. The inner region $\sum_i \delta_i^2 < \sigma$ is exactly the real ball

$$Q < R_n^2, \qquad R_n^2 = \sigma - \frac{\pi^2}{n}. \tag{4.5}$$

For fixed small $\sigma > 0$ and large n, these squared radii lie between $\sigma/2$ and σ.

The relevant pole is the unique solution of

$$\sum_i \big(g(T_n(v), d_n + v_i) - \mathrm{i}\phi\big) = 0. \tag{4.6}$$

It is defined and holomorphic on a common small ball in V_n. Indeed, after the common constant and linear terms in v_i are subtracted, the remaining sum is bounded by $C\sum_i |v_i|^2$; the parameter derivative of the left side has modulus at least cn. Rouché's theorem on a circle of radius $C\sum_i |v_i|^2/n$ about τ_n, followed by uniqueness and the implicit-function theorem, gives

$$T_n(0) = \tau_n, \qquad |T_n(v) - \tau_n| \leq \frac{C}{n} \sum_i |v_i|^2. \tag{4.7}$$

All scalar analytic bounds come from the original fixed bidisk.

Direct differentiation at the center gives

$$g(\tau_n, d_n) = \mathrm{i}\phi, \qquad g_t(\tau_n, d_n) = -\frac{\mathrm{i}B_n}{s}, \qquad \frac{1}{2} g_{\delta\delta}(\tau_n, d_n) = \frac{\mathrm{i}B_n h_n}{2s}.$$

Evenness in δ gives $g_{\delta\delta\delta}(t,d_n)=O(d_n)$. Writing $u=T_n-\tau_n$, Taylor expansion of (4.6), with the linear terms canceled by $\sum v_i=0$, therefore gives on the real ball

$$0=ng_t(\tau_n,d_n)u+\tfrac12 g_{\delta\delta}(\tau_n,d_n)Q+O\left(n|u|^2+|u|Q+d_n\sum_i|v_i|^3+\sum_i v_i^4\right).$$

Using (4.7), we obtain the following estimate:

$$\begin{aligned} T_n(v)&=T_{0,n}(v)+\mathcal{R}_n(v), \qquad T_{0,n}(v)=\tau_n+\frac{h_nQ}{2n},\\ |\mathcal{R}_n(v)|&\le\frac{C}{n}\left(\frac1n\sum_i|v_i|^3+\sum_i v_i^4+\frac{Q^2}{n}\right). \end{aligned} \tag{4.8}$$

In particular, since $\sum|v_i|^3\le Q^{3/2}$ and $\sum v_i^4\le Q^2$, a sufficiently small fixed σ gives

$$|\mathcal{R}_n(v)|\le\tfrac14 D_n(v), \qquad D_n(v):=\tau_n+\frac{a_nQ}{2n}\le|T_{0,n}(v)| \quad (Q<\sigma). \tag{4.9}$$

4.2 The complete residue density

Include every normalization in the numerator

$$F_n(t,v)=\frac{B(t)^{n^2/2}}{n!(2\pi)^{n-1}}\prod_i\frac{1}{\sinh g(t,d_n+v_i)}\prod_{i<k}\left[\frac{\sin((v_i-v_k)/2)}{\sinh((g(t,d_n+v_i)+g(t,d_n+v_k))/2)}\right]^2. \tag{4.10}$$

The square root of B is its physical one. The coefficient of the exact parameter pole is

$$I_{n,j}=\int_{Q<R_n^2}\frac{\rho_n(v)}{T_n(v)^{j+1}}\,\mathrm{d}v, \qquad \rho_n(v)=-\frac{2F_n(T_n(v),v)}{\sum_i g_t(T_n(v),d_n+v_i)}. \tag{4.11}$$

The minus sign comes from expansion of $(t-T_n)^{-1}$ at $t=0$. The ratio ρ_n/Δ^2 is holomorphic and nonzero near the center, where its value is

$$\rho_{n,0}=-\frac{2B_n^{\alpha_n-1/2}}{n\,n!(2\pi)^{n-1}2^{n(n-1)}s^{2\alpha_n}}. \tag{4.12}$$

Indeed, the numerator divided by Δ^2 at the center is $-\mathrm{i}B_n^{n^2/2}/[n!(2\pi)^{n-1}2^{n(n-1)}s^{n^2}]$, and the phase derivative is $-\mathrm{i}nB_n/s$.

Lemma 4.1 (Direct full-density estimate)**.** *On the real ball $Q<\sigma$,*

$$\rho_n(v)=\rho_{n,0}\Delta(v)^2e^{\ell_n(v)}, \qquad |\ell_n(v)|\le C(\sqrt{nQ}+nQ). \tag{4.13}$$

The logarithm is holomorphic through all coincidence divisors.

Proof. Divide every analytic factor by its nonzero center value before taking logarithms. The factor $B(T_n)^{n^2/2}$ costs $Cn^2|T_n-\tau_n|\le CnQ$. Also

$$\sum_i|g(T_n,d_n+v_i)-\mathrm{i}\phi|\le C\left(n|T_n-\tau_n|+d_n\sum_i|v_i|+Q\right).$$

Each phase occurs in $n-1$ pair denominators, so their total logarithmic cost is at most $C(nQ+\sqrt{nQ})$. The sine divided differences cost $C\sum_{i<k}(v_i-v_k)^2=CnQ$, and the one-body denominators have the smaller single-index bound. Finally, cancellation of linear terms in $\sum_i g_t(T_n,d_n+v_i)$ gives its ratio to $ng_t(\tau_n,d_n)$ as $1+O(Q/n)$. for the complete residue density. □

4.3 An angular estimate from the Gaussian period

Separate radius and angle in the positive real measure $\Delta(v)^2\,\mathrm{d}v$ on the zero-sum plane. Write $v = \sqrt{Q}\,\theta$, where $\sum \theta_i = 0$ and $\sum \theta_i^2 = 1$, and let ν_n be the normalized angular measure proportional to $\Delta(\theta)^2$.

Lemma 4.2 (Uniform angular exponential moment)**.** *For $Z_n(\theta) = n \max_i \theta_i^2$, there are absolute $c_0, C_0 > 0$ such that*

$$\sup_{n\ge 3} \mathbb{E}_{\nu_n}\big[(1+Z_n)e^{c_0 Z_n}\big] \le C_0. \tag{4.14}$$

One may take $c_0 = 1/32$.

Proof. Use the Gaussian probability measure

$$\frac{1}{\mathcal{G}_n} e^{-Q(v)} \Delta(v)^2\,\mathrm{d}v, \qquad \mathcal{G}_n = \frac{n!\pi^{(n-1)/2}P_n}{\sqrt{n}}, \qquad P_n = \prod_{r=1}^{n-1} \frac{r!}{2^r},$$

whose normalization is (3.9). Homogeneity shows that Q and θ are independent, θ has law ν_n, and Q has the gamma density $q^{\alpha_n - 1}e^{-q}/\Gamma(\alpha_n)$, where $\alpha_n = (n^2-1)/2$.

We first bound the one-coordinate marginal density $f_n(x)$. Fix $v_1 = x$, put $m = n-1$ and $c = n/(n-1)$, and write

$$v_{i+1} = u_i - x/m \quad (1 \le i \le m), \qquad \sum_i u_i = 0.$$

In the actual independent coordinates $(x, u_1, \dots, u_{m-1})$, the Jacobian is one. Moreover

$$Q(v) = cx^2 + Q(u), \qquad \Delta(v)^2 = \Delta(u)^2 \prod_{i=1}^{m} (cx - u_i)^2.$$

Consequently, with $\mathbb{E}_m$ denoting the analogous constrained Gaussian measure in m variables,

$$f_n(x) = \frac{\mathcal{G}_{n-1}}{\mathcal{G}_n} e^{-cx^2} \mathbb{E}_m \prod_{i=1}^{m} (cx - u_i)^2, \qquad \frac{\mathcal{G}_{n-1}}{\mathcal{G}_n} = \frac{2^{n-1}}{\sqrt{\pi}\,n!}\sqrt{\frac{n}{n-1}}. \tag{4.15}$$

The arithmetic–geometric mean inequality and $\sum_i u_i = 0$ give

$$\prod_{i=1}^{m} (cx - u_i)^2 \le \left(c^2x^2 + \frac{Q(u)}{m}\right)^m.$$

Here $Q(u)$ is gamma with shape $\alpha_m = (m^2-1)/2$. For $0 \le k \le m$ its moments satisfy

$$\mathbb{E}_m \left(\frac{Q(u)}{m}\right)^k = \frac{\alpha_m(\alpha_m+1)\cdots(\alpha_m+k-1)}{m^k} \le n^k,$$

because $(\alpha_m + m - 1)/m \le n$. Since $c^2 \le 9/4 < 3$, the binomial expansion therefore yields

$$\mathbb{E}_m \prod_{i=1}^{m} (cx - u_i)^2 \le (3x^2 + n)^{n-1}. \tag{4.16}$$

Use $n! \geq (n/e)^n$, $c \geq 1$, and $\sqrt{n/(n-1)}/(2\sqrt{\pi}) < 1$. With $t = x^2/n$, (4.15)–(4.16) give

$$f_n(x) \leq \frac{1}{n}(2e)^n(1+3t)^{n-1}e^{-x^2}.$$

For $t \geq 16$, $\log(2e) + \log(1+3t) \leq t/2$: it holds at 16, and the derivative of the right side minus the left side is positive thereafter. Thus

$$f_n(x) \leq \frac{1}{n}e^{-x^2/2} \qquad (x^2 \geq 16n). \tag{4.17}$$

The Gaussian measure is invariant under coordinate permutations; the independent-coordinate measure is a constant multiple of Euclidean measure on the zero-sum plane. Hence a union bound and (4.17) imply, for $A^2 \geq 16n$,

$$\Pr\left(\max_i v_i^2 \geq A^2\right) \leq 2\int_A^\infty e^{-x^2/2}\,\mathrm{d}x \leq \frac{2}{A}e^{-A^2/2}. \tag{4.18}$$

To pass to the angular variable, the inverse gamma moment gives

$$\Pr\left(Q < \frac{n^2}{8}\right) \leq \frac{n^2}{8}\mathbb{E}\frac{1}{Q} = \frac{n^2}{4(n^2-3)} \leq \frac{3}{8} \qquad (n \geq 3).$$

Independence of radius and angle, and $\max_i v_i^2 = QZ_n/n$, consequently give

$$\frac{5}{8}\Pr(Z_n \geq z) \leq \Pr\left(\max_i v_i^2 \geq \frac{nz}{8}\right).$$

For $z \geq 128$, apply (4.18) with $A^2 = nz/8 \geq 16n$ and $A \geq 1$ to conclude

$$\Pr(Z_n \geq z) \leq \frac{16}{5}e^{-nz/16}, \qquad z \geq 128. \tag{4.19}$$

Integrating this tail proves (4.14), for example with $c_0 = 1/32$, uniformly in $n \geq 3$. □

The precise consequences we need are

$$\sum_i v_i^4 \leq \frac{Q^2 Z_n}{n}, \qquad \sum_i |v_i|^3 \leq \sqrt{Q\sum_i v_i^4} \leq \frac{Q^{3/2}\sqrt{Z_n}}{\sqrt{n}}. \tag{4.20}$$

These inequalities are pointwise; the exponential moment handles all angular directions, including those where Z_n is large.

4.4 The exact complex quadratic period

Write

$$P_n = \prod_{r=1}^{n-1}\frac{r!}{2^r}, \qquad \mathcal{G}_n = \int_{V_n\cap\mathbb{R}^n} e^{-Q}\Delta(v)^2\,\mathrm{d}v = \frac{n!\pi^{(n-1)/2}P_n}{\sqrt{n}}.$$

This is the constrained Gaussian period (3.9), in the actual independent-coordinate measure. Homogeneity gives radial measure $\mathcal{G}_n Q^{\alpha_n - 1}\,\mathrm{d}Q/\Gamma(\alpha_n)$. For $j+1 > \alpha_n$ and $\Re h > 0$, therefore,

$$J_h := \int_{V_n\cap\mathbb{R}^n} \frac{\Delta(v)^2\,\mathrm{d}v}{(\tau_n + hQ/(2n))^{j+1}} = \mathcal{G}_n(2n)^{\alpha_n}h^{-\alpha_n}\tau_n^{\alpha_n-j-1}\frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)}. \tag{4.21}$$

One proof uses the scalar Laplace formula for the negative power, followed by the radial gamma integral; absolute convergence follows from $\Re h > 0$. All integrations remain over the original real variables. The gamma ratio in (4.21) is exact.

Use $a_n = \Re h_n$ for the positive majorant. The corresponding period satisfies

$$\frac{J_{a_n}}{|J_{h_n}|} = \left(\frac{|h_n|}{a_n}\right)^{\alpha_n} = \exp(O(n^{-2})). \tag{4.22}$$

Indeed, $a_n = 1 + O(n^{-2})$ and $\Im h_n = O(n^{-2})$, so the logarithm in parentheses is $O(n^{-4})$. This comparison bounds the complex quadratic period by its positive majorant on the same real integration cycle.

4.5 Uniform comparison and the two first windings

We need uniformity only over

$$\log j < n < \frac{\sqrt{j}}{(\log j)^{1/4}}. \tag{4.23}$$

In this window $n \to \infty$ and $\varepsilon_n := n^2/j \to 0$ uniformly. Let $I_{n,j}$ be the exact first-winding pole coefficient in (4.11).

Proposition 4.3 (Uniform estimate on the original real cycle)**.** *Uniformly for the odd multiples of p in* (4.23)*,*

$$I_{n,j} = \rho_{n,0} J_{h_n} \left(1 + O\left(\frac{n}{\sqrt{j}}\right)\right). \tag{4.24}$$

The sum of the two first-winding pole coefficients is consequently

$$M(n,j) \left(1 + O\left(\frac{n}{\sqrt{j}}\right)\right), \tag{4.25}$$

where the complete exact model is

$$\boxed{M(n,j) = -\frac{2^{2-\alpha_n} B_n^{\alpha_n - 1/2} P_n n^{\alpha_n - 3/2}}{\pi^{(n-1)/2} s^{2\alpha_n}} h_n^{-\alpha_n} \tau_n^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)}.} \tag{4.26}$$

Proof. Normalize the positive measure associated with J_{a_n} and set

$$x = \frac{a_n Q}{2n\tau_n}, \qquad \varepsilon = \frac{n^2}{j}.$$

Radius and angle are independent. The angular law is ν_n, and the radial density is proportional to $x^{\alpha_n - 1}(1+x)^{-j-1}$. Since $2n^2\tau_n$ and a_n are bounded above and below, $Q \asymp x/n$ uniformly. Equations (4.8) and (4.20) give

$$(j+1)\frac{|\mathcal{R}_n|}{D_n} \le C \frac{j}{n^2} \frac{x^{3/2}\sqrt{Z_n} + x^2(Z_n + 1)}{1 + x}. \tag{4.27}$$

Moreover (4.9) implies

$$\left| \frac{e^{\ell_n}}{T_n^{j+1}} - \frac{1}{T_{0,n}^{j+1}} \right| \le D_n^{-j-1} \left[\exp\left\{ |\ell_n| + \frac{2(j+1)|\mathcal{R}_n|}{D_n} \right\} - 1 \right]. \tag{4.28}$$

This follows by the convergent logarithm of $1+\mathcal{R}_n/T_{0,n}$, whose modulus is at most $2|\mathcal{R}_n|/D_n$.

Fix $K=32$. On $x\le K\varepsilon$, (4.13) and (4.27) bound the exponent in (4.28) by

$$C\{\sqrt{\varepsilon}(1+\sqrt{Z_n})+\varepsilon(Z_n+1)\}\le C'\sqrt{\varepsilon}(1+Z_n).$$

For sufficiently large j, Lemma 4.2 and the mean-value theorem for the exponential show that its angular expectation, after subtracting one, is $O(\sqrt{\varepsilon})$. This estimates the whole angular measure in the central radial region.

On the actual ball, $x\le Ln$ for a fixed L. The deterministic bound $Z_n\le n$ in (4.27) gives

$$\frac{2(j+1)|\mathcal{R}_n|}{D_n}\le Cj\frac{x^{3/2}/n^{3/2}+x^2/n}{1+x}.$$

Divided by $j\log(1+x)$, this tends to zero uniformly for $0<x\le Ln$: splitting at 1 and $\sqrt{n}$ bounds the ratio by $O(n^{-1})$, $O(n^{-1/2})$, and $O(1/\log n)$, respectively. Also $|\ell_n|\le C(\sqrt{x}+x)$. On $K\varepsilon\le x\le Ln$,

$$\frac{\sqrt{x}+x}{j\log(1+x)}=o(1)$$

uniformly. For $x\le 1$, use $j\sqrt{x}\ge\sqrt{K}\,n\sqrt{j}$; for $x\ge 1$, use $x\le Ln$ and $n/j\to 0$. Hence for all sufficiently large j in the stated window,

$$|\ell_n|+\frac{2(j+1)|\mathcal{R}_n|}{D_n}\le\frac{j}{4}\log(1+x)\quad(K\varepsilon\le x\le Ln).\tag{4.29}$$

Write $Z_\beta(\alpha,J)=\int_0^\infty x^{\alpha-1}(1+x)^{-J-1}\,\mathrm{d}x$. The normalized contribution of this remaining radial region is therefore bounded by a constant times

$$(1+K\varepsilon)^{-j/4}\frac{Z_\beta(\alpha_n,j/2)}{Z_\beta(\alpha_n,j)}\le(1+K\varepsilon)^{-j/4}4^{\alpha_n}\le e^{-cn^2}.\tag{4.30}$$

Indeed, $\alpha_n\le j/4$ eventually, and its integrality gives the exact ratio

$$\frac{Z_\beta(\alpha_n,j/2)}{Z_\beta(\alpha_n,j)}=\prod_{r=0}^{\alpha_n-1}\frac{j-r}{j/2-r}\le 4^{\alpha_n}.$$

Use $\log(1+K\varepsilon)\ge K\varepsilon/2$ to obtain the last inequality. The omitted part of the unperturbed period outside the actual ball has the same bound: its positive majorant has no perturbation factor, and the ball endpoint exceeds $K\varepsilon$ for large j.

Combining these estimates and (4.22) gives

$$\frac{|I_{n,j}-\rho_{n,0}J_{h_n}|}{|\rho_{n,0}J_{h_n}|}\le C\left(\frac{n}{\sqrt{j}}+e^{-cn^2}\right).$$

Because $n>\log j$, the exponential term is absorbed into $n/\sqrt{j}$, uniformly. This proves (4.24).

Reflection $\delta\mapsto-\delta$ preserves the full integrand and independent-angle measure, and exchanges the two first-winding balls. They therefore contribute equally. Multiplication of (4.12) by the period (4.21) and this factor two gives (4.26) exactly. □

The factor $B_n^{\alpha_n-1/2}$ carries the particle-dependent phase of the model. The comparison below uses the relative estimate on (4.23) and absolute bounds outside it.

5 Negative high-temperature edges: boundary jets and the complete odd sum

We prove nonanalyticity at negative edges through the growth of the Taylor coefficients defined by inward boundary limits.

5.1 Setting and the existence of the boundary jets

Fix a regular negative endpoint. After complex conjugation if necessary, the local analytic data can be written

$$W(t,\pi+\delta)=b+B(t)\big[t-(1-\cos\delta)\big],\qquad b=\cos\phi,\quad \phi=\frac{2\pi r}{p}\in(0,\pi),\tag{5.1}$$

where $p\ge 3$ is odd, $(r,p)=1$, B is holomorphic and nonzero, $\Im B(0)<0$, and t is a nonramified local temperature coordinate. The physical inward direction is $t=-\varepsilon$, $\varepsilon>0$. The branch $\cosh\gamma(q;t)=W(t,q)$ has $\gamma(\pi;0)=\mathrm{i}\phi$. All assertions below concern this branch. At a regular negative edge, the earlier root construction gives a common branchfree parameter neighborhood for the entire real angular circle.

Write the complete coefficient as

$$C_n(t)=\int_{\mathcal{T}_n} H_n(t,q)\coth\!\Big(\frac{\Gamma_n(t,q)}{2}\Big)\,\mathrm{d}q,\qquad \Gamma_n=\sum_{i=1}^{n}\gamma(q_i;t),\tag{5.2}$$

where $\mathcal{T}_n$ is the zero-total-momentum torus, $q_n=-\sum_{i<n}q_i$ modulo 2π, the measure $\mathrm{d}q$ is the ordinary measure in the $n-1$ independent angles, and

$$H_n(t,q)=\frac{B(t)^{n^2/2}}{n!(2\pi)^{n-1}}\prod_{i=1}^{n}\frac{1}{\sinh\gamma(q_i;t)}\prod_{i<k}\Big[\frac{\sin((q_i-q_k)/2)}{\sinh((\gamma(q_i;t)+\gamma(q_k;t))/2)}\Big]^2.\tag{5.3}$$

The branch of $B^{1/2}$ is the one in the physical representation. The normalization in (5.3) is retained throughout this section. The physical susceptibility differs from the complete odd sum by the analytic prefactor already identified in the physical representation; that prefactor is nonzero at a regular endpoint.

Set

$$S(q)=\sum_{i=1}^{n}\operatorname{dist}(q_i,\pi)^2.$$

The spectral geometry and the matched Schur–Pfaffian estimate imply, with constants independent of n,

$$\Re\gamma(q;0)\ge c_0\operatorname{dist}(q,\pi)^2,\quad |\gamma(q;0)-\mathrm{i}\phi|\le C\operatorname{dist}(q,\pi)^2,\tag{5.4}$$

$$\sup_{|t|<r_0,\ q\in\mathcal{T}_n}|H_n(t,q)|\le E_n,\qquad E_n=\exp\{-\kappa n^2+Cn\log(n+1)+C\}.\tag{5.5}$$

Choose r_0 small enough that the strict Schur bound persists by continuity on the fixed complex disk, including parameters at which $|e^{-\gamma}|>1$.

For odd n, choose the shortest lifts $q_i=\pi+\delta_i$ modulo 2π. Momentum conservation gives $\sum_i\delta_i=w\pi$ with w odd. Therefore

$$S(q)\ge\frac{\pi^2}{n}.\tag{5.6}$$

This is the source of an n^{-2} coefficientwise analytic disk, rather than a common disk for the entire particle sum.

Lemma 5.1 (Coefficientwise disks and boundary derivatives)**.** *There is $c > 0$ such that each odd coefficient is holomorphic on $|t| < c/n^2$, agrees there with* (5.2)*, and obeys*

$$\sup_{|t|<c/n^2} |C_n(t)| \le CnE_n, \qquad |[t^j]C_n(t)| \le (Cn^2)^j CnE_n. \tag{5.7}$$

The complete odd sum on $t < 0$ has derivatives of every fixed order at $t = 0$ from that side, and its normalized boundary jets are

$$a_j = \lim_{\varepsilon\downarrow 0} \frac{1}{j!}\partial_t^j \sum_{n \text{ odd}} C_n(-\varepsilon) = \sum_{n \text{ odd}} [t^j]C_n(t). \tag{5.8}$$

The sum on the right converges absolutely for each fixed j.

Proof. The parameter derivative of each one-particle phase is bounded on a fixed disk. Thus, using (5.4) and (5.6), a sufficiently small perturbation $|t| < c/n^2$ preserves $\Re\Gamma_n \ge c'/n$. The modulus of the hyperbolic cotangent is then at most Cn. This proves the first bound; Cauchy's estimate proves the second.

For the inward real approach, (5.1) gives

$$\Re\gamma(q; -\varepsilon) \ge c(\varepsilon + \operatorname{dist}(q, \pi)^2).$$

Indeed $\Im W = (-\Im B(-\varepsilon))(\varepsilon + 1 - \cos\delta)$ is positive, and writing $\gamma = v + \mathrm{i}\psi$ expresses it as $\sinh v \sin\psi$, with ψ in a compact subinterval of $(0, \pi)$. A disk centered at $-\varepsilon$ of radius $c(\varepsilon + n^{-2})$ consequently preserves the same denominator gap, up to a fixed factor. Cauchy's estimate there bounds the jth derivative by $j!(Cn^2)^j CnE_n$, uniformly for small ε. This is summable for every fixed j. Differentiation, the boundary limit, and the particle sum therefore commute by dominated convergence. □

5.2 Analytic remainder estimates

Choose $\sigma > 0$ sufficiently small, and then a fixed $c_{\text{out}} > 0$ such that $C_\Gamma c_{\text{out}} < c_0\sigma/2$ and $C_\Gamma(\sigma + c_{\text{out}}) < \pi/p$. Here C_Γ bounds the phase increments below. On $|t| < c_{\text{out}}/n$,

$$\Re\Gamma_n \ge c_0 S - C_\Gamma c_{\text{out}}, \qquad |\Gamma_n - ni\phi| \le C_\Gamma(S + c_{\text{out}}). \tag{5.9}$$

If $S \ge \sigma$, the first estimate supplies a fixed positive real part by this choice of c_{out}. If $S < \sigma$ and $p \nmid n$, the second estimate supplies a fixed phase gap: the distance from $n\phi$ to $2\pi\mathbb{Z}$ is at least $2\pi/p$. Thus the whole coefficient for $p \nmid n$, and the part with $S \ge \sigma$ for any n, are holomorphic on this c_{out}/n disk with norm at most CE_n. The angular cutoff is fixed and independent of t.

For $p \mid n$, put $D_n = \Gamma_n - ni\phi$. On the inner region $S < \sigma$, $|D_n| < \pi$ on the same disk if the constants are sufficiently small. The identity

$$\coth(D_n/2) = \frac{2}{D_n} + R(D_n), \qquad R \text{ holomorphic on } |D_n| < 2\pi, \tag{5.10}$$

places the R part in the same analytic remainder class.

For the exact parameter pole, use the unique small lifts $q_i = \pi + \delta_i$ in the inner region and put $v_i = 1 - \cos\delta_i$. Let

$$d_n(t,q) = \frac{1}{n} D_n(t,q), \qquad a_0 = \frac{B(0)}{\sinh(\mathrm{i}\phi)} \neq 0.$$

On a sufficiently small fixed parameter disk,

$$\partial_t d_n(t,q) = a_0 + O(|t| + \sigma),$$

uniformly in n and in $S < \sigma$. Reducing the disk and σ makes this derivative lie within $|a_0|/4$ of a_0. The quantitative inverse-function estimate gives a unique zero $t = T_n(q) = O(S/n)$ and the factorization

$$D_n(t,q) = n(t - T_n(q))A_n(t,q), \qquad A_n(t,q) = \int_0^1 \partial_t d_n(T_n(q) + u(t - T_n(q)), q)\,\mathrm{d}u. \tag{5.11}$$

The factor A_n is analytic and uniformly bounded away from zero on a smaller fixed disk containing every such $T_n(q)$. The proof can also be obtained directly by Rouché's theorem on a circle of radius CS/n: $|d_n(0,q)| \leq CS/n$, while the derivative differs uniformly from the nonzero constant a_0.

Set $F_n = H_n/A_n$. The exact identity

$$\frac{2H_n(t,q)}{D_n(t,q)} = \frac{2F_n(T_n(q),q)}{n(t - T_n(q))} + \frac{2}{n}\frac{F_n(t,q) - F_n(T_n(q),q)}{t - T_n(q)} \tag{5.12}$$

has a holomorphic second term, including at $t = T_n(q)$. A fixed-disk Cauchy bound on $\partial_t F_n$ bounds this divided difference by CE_n. In particular no lower bound on the distance between the pole and a chosen integration contour is being assumed.

Lemma 5.2 (Exact reduction to the inner pole integrals)**.** *For every $j \geq 0$ the boundary coefficient in* (5.8) *is the sum of the exact inner pole coefficients for odd n divisible by p, together with a remainder r_j satisfying*

$$|r_j| \leq \exp\left\{\frac{1}{2} j \log(j+1) + Cj + C\right\}. \tag{5.13}$$

For a resonant n, the inner pole coefficient is exactly

$$a_{j,n}^{\mathrm{pole}} = -\int_{S<\sigma} \frac{2H_n(T_n(q),q)}{\partial_t \Gamma_n(T_n(q),q)} T_n(q)^{-j-1}\,\mathrm{d}q. \tag{5.14}$$

Proof. All removed terms are bounded on a disk c_{rem}/n, where $c_{\mathrm{rem}} > 0$ is a fixed sufficiently small fraction of c_{out}. Construct the exact pole subtraction on its fixed disk and restrict the identity to the c/n^2 disk of the original integral. Taking Taylor coefficients there coefficients gives (5.14); the minus sign is the coefficient of $(t - T)^{-1}$ at $t = 0$. Every removed term is analytic on $|t| < c_{\mathrm{rem}}/n$ with norm bounded by CE_n. Their total coefficient bound is

$$\sum_{n \geq 1} (Cn)^j E_n.$$

Absorb $Cn\log(n+1)$ into $(\kappa/2)n^2 + C'$, and optimize $j\log n - (\kappa/2)n^2$. This gives (5.13), including summability of the tail. □

5.3 Uniform bounds for the winding sectors

The exact pole is quantitatively close to the average $\bar{v} = n^{-1}\sum_i v_i$. At $t = \bar{v}$, the linear terms in the analytic expansion of $\operatorname{arcosh}(b + B(\bar{v})(\bar{v} - v_i))$ sum to zero. The same uniform inverse-function estimate as above consequently gives

$$T_n(q) = \bar{v} + O(\mathrm{Var}(v)), \qquad |T_n(q)| \geq (1 - C\sigma)\frac{S(q)}{2n}. \tag{5.15}$$

Here $\mathrm{Var}(v) = n^{-1}\sum_i (v_i - \bar{v})^2 \leq C\sigma\bar{v}$, and $\bar{v} \geq (1 - C\sigma)S/(2n)$, giving the lower bound for the complex pole modulus.

Write $s = \sin\phi > 0$, $\alpha_n = (n^2 - 1)/2$, and

$$H_{0,n} = \frac{|B(0)|^{n^2/2}}{n!(2\pi)^{n-1}2^{n(n-1)}s^{n^2}}. \tag{5.16}$$

The complete numerator, evaluated at the pole, satisfies

$$|H_n(T_n(q), q)| \leq H_{0,n}\Delta(\delta)^2 e^{CnS}, \qquad \Delta(\delta) = \prod_{i<k}(\delta_i - \delta_k). \tag{5.17}$$

Divide each sine numerator by $(\delta_i - \delta_k)/2$ and each hyperbolic denominator by $\sinh(\mathrm{i}\phi)$. Analytic logarithms of the resulting near-one ratios are bounded by $C(\delta_i - \delta_k)^2$ and $C(|T_n| + \delta_i^2 + \delta_k^2)$, respectively. Their sums are $O(nS + n^2|T_n|) = O(nS)$. The one-body factors cost $O(S)$. Finally the logarithm of $B(T_n)^{n^2/2}/B(0)^{n^2/2}$ costs $O(n^2|T_n|)$. This proves (5.17) with the center scale (5.16).

In winding sector w, where w is odd, put

$$\delta_i = \frac{w\pi}{n} + \eta_i, \quad \sum_i \eta_i = 0, \quad Q = \sum_i \eta_i^2, \quad m_{n,w} = \frac{\pi^2 w^2}{n}.$$

Then $S = Q + m_{n,w}$ and $\Delta(\delta) = \Delta(\eta)$. There are at most $C\sqrt{n}$ sectors with nonempty $S < \sigma$. The constrained Gaussian integral, with the same independent-angle measure as above, is

$$I_n = \int_{\mathbb{R}^{n-1}} e^{-Q}\Delta(\eta)^2\,\mathrm{d}\eta = \frac{\pi^{(n-1)/2}2^{-n(n-1)/2}}{\sqrt{n}}\prod_{r=1}^{n} r!. \tag{5.18}$$

For example, separating the center of mass in the ordinary Gaussian Vandermonde integral gives a Jacobian n and a center Gaussian $\sqrt{\pi/n}$, whose product explains the constrained $1/\sqrt{n}$ factor. Homogeneity therefore gives, whenever $j + 1 > \alpha_n$,

$$\int_{\mathbb{R}^{n-1}} \frac{\Delta(\eta)^2}{(Q + m)^{j+1}}\,\mathrm{d}\eta = I_n m^{\alpha_n - j - 1}\frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j + 1)}. \tag{5.19}$$

Lemma 5.3 (Higher windings and large particle numbers)**.** *For $j + 1 > \alpha_n$, the exact pole coefficient in winding w satisfies*

$$|a^{\mathrm{pole}}_{j,n,w}| \leq \frac{CH_{0,n}}{n}e^{Cn\sigma}\left(\frac{2n}{1 - C\sigma}\right)^{j+1} I_n m_{n,w}^{\alpha_n - j - 1}\frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j + 1)}. \tag{5.20}$$

For every resonant odd n and every $j \geq 0$, without a restriction involving α_n,

$$\log\sum_w |a^{\mathrm{pole}}_{j,n,w}| \leq 2j\log n - \frac{1}{2}n^2\log n + Cn^2 + Cj + C. \tag{5.21}$$

Empty sectors are understood to contribute zero.

Proof. The residue denominator has modulus at least cn. Combine this with (5.15) and (5.17), replace S in its exponential by σ, and extend the positive radial majorant to the whole real zero-sum plane. Equation (5.19) proves (5.20). Its exact dependence on w relative to the same expression with $w = 1$ is $|w|^{-2(j+1-\alpha_n)}$.

For the second estimate, use the simpler bound $|T_n| \geq c/n^2$ and the exact ball mass

$$\int_{Q<\sigma} \Delta(\eta)^2 \, \mathrm{d}\eta = I_n \frac{\sigma^{\alpha_n}}{\Gamma(\alpha_n + 1)}.$$

The logarithm of this ball mass is $-\frac{1}{2}n^2 \log n + O(n^2)$; this follows by applying Stirling's inequalities to (5.18) and to $\Gamma(\alpha_n + 1)$. Multiplication by $H_{0,n}$ costs $O(n^2)$. The number of windings and the remaining polynomial factors are absorbed into the displayed error. This proves (5.21) for arbitrarily large n. □

5.4 The exact first-winding model

Let

$$\tau_n = 1 - \cos(\pi/n), \quad B_n = B(\tau_n), \quad h_n = \cos(\pi/n) - \frac{B_n b}{s^2} \sin^2(\pi/n), \quad P_n = \prod_{r=1}^{n-1} r!\, 2^{-r}. \tag{5.22}$$

In particular $h_n = 1 + O(n^{-2})$. Proposition 4.3 in Section 4 gives, uniformly for odd multiples of p in the window

$$\log j < n < \frac{\sqrt{j}}{(\log j)^{1/4}},$$

$$a^{\mathrm{pole}}_{j,n,1} + a^{\mathrm{pole}}_{j,n,-1} = M(n,j)(1 + e_{n,j}), \tag{5.23}$$

where

$$M(n,j) = -\frac{2^{2-\alpha_n} B_n^{\alpha_n - 1/2} P_n n^{\alpha_n - 3/2}}{\pi^{(n-1)/2} s^{2\alpha_n}} h_n^{-\alpha_n} \tau_n^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)}, \tag{5.24}$$

and

$$|e_{n,j}| \leq C \frac{n}{\sqrt{j}}. \tag{5.25}$$

The estimate applies to the ball of squared radius $\sigma - \pi^2/n$ on the original real angular cycle.

The constants in (5.24) follow from the central numerator, pole derivative and constrained beta period. At the collision $\gamma_i = \mathrm{i}\phi$, the numerator divided by Δ^2 is

$$-\frac{\mathrm{i} B_n^{n^2/2}}{n!(2\pi)^{n-1} 2^{n(n-1)} s^{n^2}}, \qquad \Gamma_t = -\frac{\mathrm{i} n B_n}{s}.$$

Multiplication by $-2/\Gamma_t$, followed by the exact complex quadratic beta integral, gives half of (5.24). Reflection $\delta \mapsto -\delta$ preserves the full integrand and exchanges the first two winding sectors, giving the displayed factor of two. The factor $B_n^{\alpha_n - 1/2}$ retains its quadratic particle-dependent phase. No positivity across particle indices is being assumed.

5.5 A discrete concavity estimate with controlled errors

Write $L(n,j) = \log|M(n,j)|$ and let $d = 2p$ be the step between successive allowed odd multiples of p. The model gives the leading location of the maximum through

$$L(n,j) = 2j\log n - \frac{n^2}{2}\log j - j\log(\pi^2/2) + O\left(n^2 + \frac{n^4}{j} + \frac{j}{n^2}\right), \tag{5.26}$$

when $n \to \infty$ and $n^2/j \to 0$. The finite differences below use the exact factors.

Lemma 5.4 (Exact-factor differences)**.** *For integer $j > \alpha_{n+d}$, uniformly as $n \to \infty$ and $n^2/j \to 0$,*

$$L(n+d,j) - 2L(n,j) + L(n-d,j) = -d^2\left(\frac{2j}{n^2} + \log j\right) + O_d\left(1 + \frac{n^2}{j} + \frac{j}{n^4}\right), \tag{5.27}$$

$$\frac{L(n+d,j) - L(n-d,j)}{2d} = \frac{2j}{n} - n\log j + O_d\left(n + \frac{n^3}{j} + \frac{j}{n^3}\right). \tag{5.28}$$

In particular the second difference is at most $-cd^2\log j$ throughout

$$\log j < n < \frac{\sqrt{j}}{(\log j)^{1/4}}, \tag{5.29}$$

whenever both neighboring indices are in this interval, for all sufficiently large j.

Proof. Put $A = \pi^2/2$ and $e_n = \log(\tau_n n^2/A)$. The elementary expansions of τ_n, $B(\tau_n)$ and h_n are convergent in n^{-2} near zero. Consequently the exact logarithm of (5.24) can be organized as

$$\begin{aligned} L(n,j) = {} & \log P_n + (2j - n^2/2 + 1)\log n - \alpha_n \log j - j\log A \\ & + R_n - je_n + E_j(\alpha_n), \\ E_j(a) = {} & \sum_{r=0}^{a-1} -\log(1 - r/j). \end{aligned} \tag{5.30}$$

Here $e_n = O(n^{-2})$ has such a convergent expansion, while R_n is a quadratic polynomial in n plus a convergent series $O(n^{-2})$. The quadratic coefficients depend on $B(0)$ and s. This follows by grouping the powers of n and $\tau_n = An^{-2}e^{e_n}$. The remaining factors $B(\tau_n)$ and h_n are nonzero near $n^{-1} = 0$. Thus R_n has centered first difference $O_d(n)$ and second difference $O_d(1)$, while the corresponding differences of je_n are $O_d(j/n^3)$ and $O_d(j/n^4)$.

The factorial products have the exact quotients

$$\log P_{n+d} - \log P_{n-d} = \sum_{r=n-d}^{n+d-1} \big(\log(r!) - r\log 2\big),$$

$$\log P_{n+d} - 2\log P_n + \log P_{n-d} = \sum_{k=0}^{d-1}\sum_{\ell=1}^{d} \log(n - d + k + \ell) - d^2\log 2.$$

Stirling's inequality applied to the finitely many factorials in the first expression gives $2dn\log n + O_d(n)$; the second expression is $d^2\log n + O_d(1)$. Taking differences of the explicit elementary function $(2j - n^2/2 + 1)\log n$ cancels these $\log n$ terms and leaves respectively $2j/n + O_d(n + j/n^3)$ after division by $2d$, and $-2d^2 j/n^2 + O_d(1 + j/n^4)$.

It remains to control E_j. On $0 \le r \le \alpha_{n+d}$,

$$0 \le -\log(1-r/j) \le Cn^2/j, \qquad \frac{\mathrm{d}}{\mathrm{d}r}[-\log(1-r/j)] \le C/j.$$

The difference $E_j(\alpha_{n+d}) - E_j(\alpha_{n-d})$ has $2dn$ summands, so it is $O_d(n^3/j)$. For the second difference, the two increments of α_n have lengths $dn + d^2/2$ and $dn - d^2/2$. The d^2 excess summands cost $O_d(n^2/j)$. Pair each remaining upper summand with its lower counterpart: their arguments differ by $O_d(n)$, so the derivative bound gives another total $O_d(n^2/j)$. Finally the differences of $-\alpha_n \log j$ are exactly $-n \log j$ after division by $2d$, and $-d^2 \log j$. This proves both stated formulas.

Stirling's bounds give $\log P_n = \frac{1}{2}n^2 \log n + O(n^2)$, and $E_j(\alpha_n) = O(n^4/j)$. Substitution into (5.30) proves (5.26). In (5.29), $n^2/j = o(1)$ uniformly, and j/n^4 is smaller than j/n^2 by at least the factor $(\log j)^{-2}$. The negative second-difference bound therefore holds throughout the stated interval. □

5.6 Selecting an integer order with one dominating index

Fix a large allowed particle index n, and for integer $j > \alpha_{n+d}$ put $F_n(j) = L(n+d, j) - L(n-d, j)$. The gamma factors in the exact model satisfy a one-step recurrence, giving the exact identity

$$F_n(j+1) - F_n(j) = \log \frac{\tau_{n-d}}{\tau_{n+d}} + \log \frac{j+1-\alpha_{n+d}}{j+1-\alpha_{n-d}}. \tag{5.31}$$

Uniformly for $\frac{1}{2}n^2 \log n \le j \le \frac{3}{2}n^2 \log n$, this increment is $4d/n + O_d(1/(n \log n))$, hence positive and $O_d(1/n)$.

Equation (5.28) proves an actual sign change on this integer interval. Indeed, at $j = cn^2 \log n + O(1)$ with $c = 1/2$ or $3/2$, its leading expression is

$$\frac{2j}{n} - n \log j = 2(c-1)n \log n - n \log \log n - n \log c + O(1/n),$$

whereas the error in (5.28) is $O_d(n)$. Thus the two endpoints have opposite signs for large n. Choose j_n as the first integer in this interval with $F_n(j_n) \ge 0$. The recurrence then gives $0 \le F_n(j_n) = O_d(1/n)$. Substituting this in (5.28), and using $\log j_n = 2 \log n + O(\log \log n)$, yields

$$j_n \sim n^2 \log n, \qquad F_n(j_n) = O_d(1/n). \tag{5.32}$$

The selected index n and its neighbors are strictly inside (5.29) for these j_n.

Let $\ell_k = L(n + kd, j_n)$ for all allowed indices in that window. Lemma 5.4 says that successive differences of ℓ_k decrease by at least $cd^2 \log j_n$. Since $\ell_1 - \ell_{-1} = F_n(j_n) = O_d(1/n)$, the two adjacent gaps satisfy $\ell_0 - \ell_{\pm 1} \ge c' d^2 \log j_n$ for large n. Summing the decreasing successive differences in both directions gives

$$\ell_k - \ell_0 \le -c'' d^2 k^2 \log j_n + O_d(|k|/n), \qquad k \ne 0.$$

Therefore

$$\sum_{\substack{m \ne n,\ p \mid m,\ m \text{ odd} \\ m \text{ in (5.29)}}} |M(m, j_n)| \le |M(n, j_n)|\, O(j_n^{-c''' d^2}). \tag{5.33}$$

No common phase of these terms is assumed. A uniform $o(1)$ relative error in the local model remains sufficient, because it multiplies the already summable moduli on the left.

5.7 Completing the all-index comparison

For $j \to \infty$, split the odd particle indices at

$$n_{\mathrm{lo}} = \log j, \qquad n_{\mathrm{hi}} = \frac{\sqrt{j}}{(\log j)^{1/4}}.$$

The low range, including every nonasymptotic index, is bounded by Lemma 5.1:

$$\sum_{n \le n_{\mathrm{lo}}} |[t^j] C_n| \le \exp\{2j \log\log j + O(j)\}. \tag{5.34}$$

All c/n analytic remainders have the smaller scale $\exp\{\frac{1}{2} j \log j + O(j)\}$ by (5.13).

For $n \ge n_{\mathrm{hi}}$, use the ball-mass bound (5.21). In the variable $x = n^2$ its controlling exponent is

$$j \log x - \tfrac{1}{4} x \log x + Cx + Cj,$$

which is decreasing for every $x \ge j/\sqrt{\log j}$ when j is sufficiently large. At the threshold it equals at most

$$j \log j - \tfrac{1}{2} j \log\log j - \tfrac{1}{4} j \sqrt{\log j} + o(j\sqrt{\log j}). \tag{5.35}$$

Its subsequent decrease makes the full infinite tail summable, with the same leading bound. This estimate handles all windings in the high range and requires no beta integral outside its convergence range.

In the intermediate range, the error in (5.23) is uniformly $O((\log j)^{-1/4}) = o(1)$. It remains to compare the higher odd windings to that model. Let $M_0(n, j)$ be the modulus of (5.24) with B_n replaced by $B(0)$, h_n by 1, and τ_n by $\pi^2/(2n^2)$. The exact formulas give

$$\left| \log \frac{M_0(n, j)}{|M(n, j)|} \right| \le C(j/n^2 + 1) = o(j) \tag{5.36}$$

uniformly in this entire range. Equation (5.20) is bounded by $CM_0(n, j)$ times

$$e^{Cn\sigma}(1 - C\sigma)^{-j-1} |w|^{-2(j+1-\alpha_n)}.$$

Here $n = o(j)$ and $\alpha_n/j = o(1)$ uniformly. Choose the fixed σ so small that $-\log(1 - C\sigma) < \frac{1}{2} \log 3$. Summing over all odd $|w| \ge 3$ then gives

$$\sum_{|w| \ge 3} |a^{\mathrm{pole}}_{j,n,w}| \le |M(n, j)| e^{-cj} \tag{5.37}$$

for some $c > 0$ and all sufficiently large j, uniformly over the intermediate range. Thus no unexamined winding or particle range remains between the preceding estimates.

At the selected sequence (5.32), the coarse expansion (5.26) suffices for the exact leading growth constant. Indeed $j_n \sim n^2 \log n$ implies

$$2 \log n = \log j_n - \log\log j_n + \log 2 + o(1), \qquad \frac{n^2 \log j_n}{2 j_n} \longrightarrow 1.$$

Every error in (5.26) is $o(j_n)$ at this scale. Consequently

$$\log |M(n, j_n)| = j_n \log j_n - j_n \log\log j_n + j_n(\log(4/\pi^2) - 1) + o(j_n), \tag{5.38}$$

$$|M(n, j_n)|^{1/j_n} \sim \frac{4 j_n}{e \pi^2 \log j_n}.$$

The exact gamma ratio was retained in the finite-difference and relative local estimates. Here its logarithmic correction to $-\alpha_n \log j_n$ is $O(n^4/j_n) = o(j_n)$, as included in (5.26).

The bounds (5.34) and (5.35), and all analytic remainders, are $o(|M(n, j_n)|)$ by (5.38). The intermediate higher windings have the same property by (5.37) and (5.33). Finally the uniform relative error in (5.23), summed over that range, is at most $o(1)\sum_m |M(m, j_n)| = o(|M(n, j_n)|)$. Therefore the complete boundary coefficient satisfies

$$a_{j_n} = M(n, j_n)(1 + o(1)). \tag{5.39}$$

This proves noncancellation in the infinite odd-particle sum even though the model phases vary with n.

Theorem 5.5 (A regular rational negative edge is singular for the full odd sum)**.** *Under the physical-branch and nonramified-coordinate hypotheses of* (5.1)*, the complete odd sum has all inward boundary derivatives at $t = 0$, but the Taylor series of those boundary derivatives has radius of convergence zero. More precisely, along the integer sequence* (5.32)*,*

$$|a_{j_n}|^{1/j_n} \sim \frac{4j_n}{e\pi^2 \log j_n} \longrightarrow \infty.$$

Consequently the complete high-temperature physical germ has no holomorphic continuation across this regular negative edge point. Multiplication by its nonzero analytic physical normalization preserves this conclusion.

Proof. The jets exist and equal the absolutely convergent coefficient sums by Lemma 5.1. Equations (5.38) and (5.39) give their stated growth and zero Taylor radius. If there were a holomorphic continuation across the point, its Taylor coefficients would be these same boundary jets and would satisfy a positive-radius Cauchy bound, a contradiction. The physical normalization can be divided out in a neighborhood because it is analytic and nonzero there. The use of the nonramified coordinate t makes this a statement about the fixed-ratio one-variable physical germ itself. □

6 Unequal-coupling boundary theorem and isotropic specialization

6.1 Completing the unequal-coupling theorem

The holomorphic sums in Section 1 agree with the physical susceptibilities on the respective real intervals. The identity theorem fixes the continuations on the connected domains Ω_H and Ω_L.

For each fixed $\lambda \neq 1$, the profile-intersection set and the two lens tips in Section 2 form a locally discrete exceptional set on the physical boundaries. Every other boundary point has a real analytic edge arc with exactly one endpoint cosine, a strictly interior second cosine, a nonzero real-linear determinant, and a nonramified holomorphic edge coordinate. Restricting that coordinate to the arc gives an open real interval. Cosines of primitive odd-prime rational angles are dense in every such interval.

On every regular LT arc, Section 3 proves a genuine singularity of the entire even-particle sum at each point of this dense family, including both endpoint signs and both transfer-axis choices. The first singular derivative has a nonzero coefficient and every higher even index is included in the absolute bounded tail. On every regular positive HT arc, the odd counterpart proves the

analogous whole-sum singularity. On every regular negative HT arc, Sections 4–5 show that all fixed boundary jets exist, but a sequence of their Taylor coefficients satisfies

$$|a_{j_n}|^{1/j_n} \sim \frac{4j_n}{e\pi^2 \log j_n} \longrightarrow \infty. \tag{6.1}$$

These coefficients are those of the complete odd sum. The Gaussian bound permits taking the boundary limit through the sum for each fixed derivative. If a holomorphic extension existed, its derivatives would have these limits and its Taylor series would have positive radius, contradicting (6.1). Division by the local analytic nonzero physical prefactor proves the same assertion for $\mathcal{X}_H$, and multiplication by the nonzero β gives χ_H.

Finally suppose a holomorphic extension existed through any boundary point, including an exceptional point. A small disk of extension contains regular arcs of the same physical boundary, and hence contains a nearby point of the appropriate dense true-singularity family. Restricting the extension to that point contradicts its already proved nonremovability. The complete graph $\partial\Omega_L$ and the complete closed curve $\partial\Omega_H$ therefore are natural boundaries. This proves the unequal-coupling part of Theorem 1.1.

For unequal couplings $\mathcal{S}_\lambda$ has interior. The natural-boundary conclusion concerns the explicitly determined boundaries of its two physical complementary components. Their nonextension follows from the complete-sum estimates and the density argument above.

6.2 Every fixed ratio and exceptional contact

For $\lambda \neq 1$, the signed-profile difference in (2.7) is a nonzero real-analytic function on the entire real axis. Its isolated zeros include all contacts of the two endpoint graphs, independently of whether the ratio is rational or irrational. Away from those contacts and the two lens tips, the endpoint-defect identity (2.9) proves strictness of the interior cosine and independence of the two spectral directions; the explicit endpoint-root derivative proves nonramification. The closure argument therefore includes every finite exceptional point. The odd-over-odd rational contacts at the imaginary axis are covered by the same analytic-profile argument.

At a selected negative HT edge, the proof uses the holomorphic nonvanishing germ $B(t)$, the exact centers $1 - \cos(\pi/n)$, the real-cycle beta estimate and the single-index comparison. These estimates apply for each fixed λ and regular point. At $\lambda = 1$ the proof in Section 7 applies directly.

6.3 Isotropic coefficient identities

Set $\lambda = 1$ and $s = \sinh z$. Then $S = T = s$, $p = s^2$, $B = 1$, and

$$D(x, y; z) = s\left[s + s^{-1} - \tfrac{1}{2}(x + x^{-1}) - \tfrac{1}{2}(y + y^{-1})\right].$$

On the positive LT interval $\sqrt{p} = s$, so every factor $\sqrt{p}/D$ in (1.10) is exactly the normalized isotropic factor. The even coefficients and the full-site identity agree term by term. In particular the function to which the isotropic result applies is

$$\mathcal{X}_L(s) = M^2 \sum_{N\geq 2,\ N \text{ even}} C_N(s) = 1 - M^2 + 2M^2 \sum_{N\geq 2,\ N \text{ even}} T_N(s), \qquad M^2 = (1 - s^{-4})^{1/4}. \tag{6.2}$$

The term $1 - M^2$ is the onsite connected correlation.

In HT, (1.13) has $W = s + s^{-1} - \cos q$ and $B = 1$. In the high-temperature interval $0 < s < 1$, the exterior parameter is $t = 1/s > 1$ and the complete angular coefficient is $C_N(s)$ as defined by (1.13). Consequently

$$\mathcal{X}_H(s) = \frac{(1 - s^4)^{1/4}}{s} \sum_{N \geq 1,\ N \text{ odd}} C_N(s). \tag{6.3}$$

The physical state and energy-field identifications in Section 1 apply to these formulas at $B = 1$.

7 The isotropic slice from its full physical product

Set $\lambda = 1$ and $s = \sinh z$. Write S for the dispersion parameter $C = s + s^{-1}$ at $B = 1$.

For $\mathsf{S} \in \mathbb{C} \setminus [-2, 2]$ and $y = e^{\mathrm{i}\theta}$, let

$$W = \mathsf{S} - \cos\theta, \qquad \rho + \rho^{-1} = 2W, \quad |\rho| < 1, \qquad \sinh\gamma = (\rho^{-1} - \rho)/2.$$

The root is unique and holomorphic: a root on the unit circle would force $W \in [-1, 1]$, and the two roots are reciprocal. Write

$$P_{ij} = -\frac{(y_i - y_j)^2 \rho_i \rho_j}{y_i y_j (1 - \rho_i \rho_j)^2}, \qquad Z = \prod_i \rho_i, \qquad a_+ = 0, \quad a_- = \pi.$$

For either sign $\sigma \in \{1, -1\}$ define the exact auxiliary coefficient

$$\mathcal{F}_N^\sigma(\mathsf{S}) = \frac{1}{N!} \int_{\mathbb{H}_N^\sigma} \frac{1 + \sigma Z}{1 - \sigma Z} \prod_i \frac{1}{\sinh\gamma_i} \prod_{i<j} P_{ij}\, d\mu_N, \qquad \mathbb{H}_N^\sigma = \left\{ \sum_i \theta_i = a_\sigma \pmod{2\pi} \right\}, \tag{7.1}$$

where

$$d\mu_N = \prod_{i<N} \frac{d\theta_i}{2\pi}, \qquad \theta_N = a_\sigma - \sum_{i<N} \theta_i \pmod{2\pi}.$$

Thus this is Euclidean hypersurface measure divided by $\sqrt{N}(2\pi)^{N-1}$. For $N = 1$ the constrained integral is evaluation at its single angle. All pair factors are rational squares, so no choice of half-angle or logarithm changes the coefficient. Put

$$C_N^\sigma(t) = \mathcal{F}_N^\sigma(t + t^{-1}), \qquad |t| > 1. \tag{7.2}$$

The compact normal-convergence bounds for the full product apply to each fixed differentiated series here. To see directly that the auxiliary sign causes no change, on a compact dispersion neighborhood all roots satisfy $|\rho_i| \leq q < 1$ and have imaginary parts of one sign. The latter fact follows from $\Im W_i = \Im\rho_i(1 - |\rho_i|^{-2})/2$. The identity

$$|1 - vw|^2 - |v - w|^2 = (1 - |v|^2)(1 - |w|^2) + 4\Im v \Im w$$

therefore bounds every root Schur factor $|(\rho_i - \rho_j)/(1 - \rho_i\rho_j)|$ strictly below one, uniformly. Expand the angular Schur product by the matching identity proved in Section 1 and cancel each matched angular pole with its corresponding root factor. The resulting complete matched pairs are bounded; all but $O(N)$ root factors retain the strict contraction. The number of matchings costs $\exp(O(N\log(N+1)))$, while the unmatched root factors supply $\exp(-cN^2)$. This gives a summable coefficient bound on the normalized constrained torus for either sign. All one-body factors and $1 - \sigma Z$ have fixed gaps there. Cauchy's inequality on a slightly larger compact neighborhood gives every fixed derivative.

7.1 Exact physical and sign identities

The physical real-temperature identities, with their analytic continuations fixed by those real values, are exactly

$$\mathcal{X}_L(s) = (1-s^{-4})^{1/4} \sum_{\substack{N\geq 2\\ N \text{ even}}} C_N^+(s), \qquad |s|>1, \tag{7.3}$$

$$\mathcal{X}_H(1/t) = Q_H(t) \sum_{\substack{N\geq 1\\ N \text{ odd}}} C_N^+(t), \qquad Q_H(t) = t(1-t^{-4})^{1/4}, \quad |t|>1. \tag{7.4}$$

The fourth roots tend to one at infinity. These identities follow by putting $p = s^2$ and $B = 1$ in the main physical formulas. In the LT identity the zero-particle term has already been subtracted in the plus state. The lattice origin is included: its contributions are $1-(1-s^{-4})^{1/4}$ in LT and 1 in HT. In particular there is no additional constant to append to either full coefficient sum.

Angle reflection preserves $\mathbb{H}_N^\sigma$, its Haar measure and each complete factor in (7.1). Conjugation also commutes with the coefficients. Replacing S, y_i, ρ_i by $-\mathsf{S}, -y_i, -\rho_i$ gives the precise negation identity

$$\mathcal{F}_N^\sigma(-\mathsf{S}) = (-1)^N \mathcal{F}_N^{(-1)^N\sigma}(\mathsf{S}). \tag{7.5}$$

Every pair is unchanged, every one-body factor changes sign, and $\prod_i y_i$ and Z both acquire $(-1)^N$. This proves (7.5) with its constraint and product-kernel signs. In particular the even physical coefficients are even in t, whereas the odd coefficients satisfy

$$C_N^+(-t) = -C_N^-(t) \qquad (N \text{ odd}). \tag{7.6}$$

The two one-particle formulas, useful below, are

$$\mathcal{F}_1^+(\mathsf{S}) = \frac{1}{\mathsf{S}-2}, \qquad \mathcal{F}_1^-(\mathsf{S}) = \frac{1}{\mathsf{S}+2}. \tag{7.7}$$

For example the second formula follows by taking $\theta = \pi$ and using $(1-\rho)/(1+\rho)\sinh(\gamma)^{-1} = 2\rho/(1+\rho)^2$.

7.2 Prime arithmetic and both HT geometries

Lemma 7.1 (The first order and the branch-mean gap)**.** *Let $p \geq 5$ be prime, $1 \leq a \leq (p-1)/2$, and $\alpha = 2\pi a/p$, $\zeta = e^{\mathrm{i}\alpha}$, $\mathsf{S}_* = 2\cos\alpha$. The first odd positive integer n admitting*

$$\mathsf{S}_* = \cos(2\pi l/n) + \cos(2\pi m/n) \tag{7.8}$$

is $n = p$. At p its unordered cosine pair is uniquely $\{\cos\alpha, \cos\alpha\}$. If $0 < \alpha < \pi/2$, the corresponding first even order is $2p$, and the same uniqueness holds there. For $b = \arccos(|\mathsf{S}_*|-1)$, $e^{-\mathrm{i}b}$ is algebraic and is not a root of unity. The selected points are dense in both open upper quadrants for odd orders, and in the first quadrant for even orders.

Proof. Write $d = (p-1)/2$, $\zeta_p = e^{2\pi\mathrm{i}/p}$ and $u_j = \zeta_p^j + \zeta_p^{-j}$ for $1 \leq j \leq d$. These d numbers are linearly independent over $\mathbb{Q}$. Indeed a relation gives a polynomial of degree at most $p-1$, with zero constant term, vanishing at ζ_p; its divisibility by $1+X+\cdots+X^{p-1}$ forces every coefficient to vanish. Each u_a is irrational, since otherwise ζ_p^a would have degree at most two over $\mathbb{Q}$.

If $p \nmid n$, the two cyclotomic fields $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_n)$ have intersection $\mathbb{Q}$: their compositum has degree $\varphi(pn) = \varphi(p)\varphi(n)$. Thus (7.8) is impossible at every odd $n < p$ and every even $n < 2p$. The equal pair realizes orders p and $2p$.

Put $\delta = \operatorname{sgn}(u_a)$ and $\mathsf{S}_+ = \delta u_a$. The conjugates of u_a are all the u_j, including a positive and a negative number, so S_+ has a negative conjugate. The number $\xi = e^{-\mathrm{i}b}$ satisfies $\xi + \xi^{-1} = 2\mathsf{S}_+ - 2$ and is algebraic. If it were a root of unity, every conjugate of the right side would lie in $[-2, 2]$. A negative conjugate of S_+ makes it strictly less than -2, a contradiction. Embeddings extend to a field containing ξ, so this conjugate argument is legitimate.

At odd order p, the cosine -1 is absent. A cosine 1 has an impossible partner when $\mathsf{S}_* < 0$, and when $\mathsf{S}_* > 0$ it would make ξ a root of unity. All remaining twice-cosines are u_j; independence applied to $2u_a = u_l + u_m$ gives uniqueness. At even order $2p$ and $\mathsf{S}_* > 0$, a cosine -1 is impossible and 1 is excluded as before. The other twice-cosines are $\pm u_j$, and the same independence again forces both to equal u_a. Finally choose arbitrarily large primes and a nearest integer to $p\alpha_0/(2\pi)$ for any prescribed angle in the stated open interval. This proves density. □

All estimates below are performed at a positive dispersion point. For LT take $0 < \alpha < \pi/2$, $\sigma = +$ and $n_* = 2p$. For an HT prime point in the first quadrant take $\sigma = +$ and $n_* = p$. At an HT prime point in the second quadrant, put

$$\widehat{\alpha} = \pi - \alpha, \qquad \widehat{\zeta} = -\overline{\zeta} = e^{\mathrm{i}\widehat{\alpha}}, \qquad \widehat{\mathsf{S}}_* = -\mathsf{S}_* > 0, \qquad \sigma = -, \quad n_* = p. \tag{7.9}$$

Then $p\widehat{\alpha}$ is an odd multiple of π. The signed equality condition has both angles in $(2\pi\mathbb{Z} + \pi)/n$. Adding π to both angles, for odd n, converts it to the ordinary condition at the negative dispersion point. Lemma 7.1 therefore proves absence of lower signed orders and uniqueness at p as well. The two equality centers in positive geometry are $\theta_i = \pm\widehat{\alpha}$. The exact derivative transfer back to the physical coefficient is

$$(C_N^+)^{(j)}((1+\varepsilon)\zeta) = (-1)^{j+1}\overline{(C_N^-)^{(j)}((1+\varepsilon)\widehat{\zeta})}. \tag{7.10}$$

This follows from (7.6) and conjugation.

In the positive geometry, write the angle as $\alpha_+ \in (0, \pi/2)$ and keep its corresponding sign σ. For every finite set of particle orders the constrained equality means

$$\theta_0 = (2\pi l + a_\sigma)/n$$

stay a positive distance from the lower branch $-b$. Otherwise $e^{-\mathrm{i}nb} = \sigma$, contrary to Lemma 7.1. The radial path and derivative budget are

$$t_\varepsilon = (1+\varepsilon)e^{\mathrm{i}\alpha_+}, \qquad \mathsf{S}_\varepsilon = t_\varepsilon + t_\varepsilon^{-1} = 2\cos\alpha_+ + 2\mathrm{i}\sin\alpha_+\varepsilon + O(\varepsilon^2), \qquad k = \left\lfloor \frac{n_*^2 - 1}{2} \right\rfloor. \tag{7.11}$$

In particular $\Re(\mathsf{S}_\varepsilon - \mathsf{S}_*) = O(\varepsilon^2)$.

7.3 One analytic tail estimate for both isotropic parities

We use the signed coefficients $\mathcal{F}_N^\sigma(\mathsf{S})$, $\sigma \in \{1, -1\}$, on $\mathbb{H}_N^\sigma = \{\sum \theta_i = a_\sigma \pmod{2\pi}\}$, where $a_+ = 0$ and $a_- = \pi$. Thus, with the independent-coordinate Haar measure $\prod_{i<N} \mathrm{d}\theta_i/(2\pi)$,

$$\mathcal{F}_N^\sigma(\mathsf{S}) = \frac{1}{N!}\int_{\mathbb{H}_N^\sigma} \frac{1+\sigma Z}{1-\sigma Z}\prod_i \frac{1}{\sinh\gamma_i}\prod_{i<l} P_{il}\prod_{i<N}\frac{\mathrm{d}\theta_i}{2\pi}, \qquad P_{il} = \frac{\sin^2((\theta_i-\theta_l)/2)}{\sinh^2((\gamma_i+\gamma_l)/2)}. \tag{7.12}$$

Here $\cosh\gamma_i = \mathsf{S} - \cos\theta_i$, $z_i = e^{-\gamma_i}$ is the interior root, and $Z = \prod_i z_i$. Fix $\mathsf{S}_* = 2\cos\alpha \in (0,2)$, and approach it with

$$\Im\mathsf{S}_\varepsilon \asymp \varepsilon > 0, \qquad \Re\mathsf{S}_\varepsilon - \mathsf{S}_* = O(\varepsilon^2). \tag{7.13}$$

All constants below may depend on this point and on fixed derivative orders, but never on N or ε. The radial exterior approach in the physical coordinate satisfies (7.13).

Theorem 7.2 (Complete isotropic higher tail)**.** *Let $n_* \geq 2$, and put $k = (n_*^2-1)/2$ when n_* is odd and $k = n_*^2/2 - 1$ when n_* is even. For either sign σ,*

$$\sum_{\substack{N\geq n_*+2\\ N\equiv n_* \ (2)}} |\partial_{\mathsf{S}}^j \mathcal{F}_N^\sigma(\mathsf{S}_\varepsilon)| \leq C_j(1+|\log\varepsilon|), \qquad 0\leq j\leq k. \tag{7.14}$$

The estimate includes every higher particle order. Under any fixed holomorphic change of the exterior parameter, it also holds for all parameter derivatives through order k.

7.3.1 A deformation of the complete constrained torus

Set $b = \arccos(\mathsf{S}_* - 1)$; then $\alpha < b < 2\alpha$. Choose a smooth periodic p, supported in $(0,\pi)$, with $0 \leq p \leq 1$ and $p = 1$ on $[\delta, \pi-\delta]$; put $m(\theta) = p(-\theta)$. The endpoint width δ is small, so both $\pm b$ lie in the corresponding plateaus. With a small fixed $\tau > 0$, define

$$P = \sum_i p(\theta_i), \quad Q = \sum_i m(\theta_i), \quad h_i = \frac{\tau}{N}(Pm(\theta_i) - Qp(\theta_i)), \quad y_i(\xi) = e^{i\theta_i + \xi h_i}, \quad 0\leq\xi\leq 1. \tag{7.15}$$

The sum of the h_i is zero, so the constrained product of the y_i is unchanged. For $y = e^{h+i\theta}$,

$$\Im\left(\mathsf{S} - \frac{y+y^{-1}}{2}\right) = \Im\mathsf{S} - \sinh h\sin\theta.$$

The motion is inward on the upper semicircle and outward on the lower one. Its imaginary part therefore remains positive throughout the homotopy. All roots stay on the interior sheet, all one-body and pair denominators stay nonzero, and $|Z| < 1$. The complete holomorphic integrand can consequently be transported over this closed constrained torus. The weights below are introduced after the homotopy.

Write $w_i = \theta_i - ih_i$ at its endpoint, and $M = I - iDh$. Because $\mathbf{1}^T M = \mathbf{1}^T$, the full determinant of M equals its tangent determinant on the constrained hyperplane. Moreover

$$Dh = \frac{\tau}{N}\operatorname{diag}(Pm_i' - Qp_i') + \frac{\tau}{N}(m\otimes p' - p\otimes m').$$

Each summand has operator norm $O(\tau)$, independently of N. Thus M^{-1} is uniformly bounded, $|\det M| \leq C^N$, and its fixed angular jets cost at most $C_j^N N^{C_j}$, by the determinant and inverse-matrix differentiation rules.

Let $\eta = 0$ on $(-\infty, 1/4]$ and $\eta = 1$ on $[1/2,\infty)$, with $0\leq\eta\leq 1$. On the final contour use

$$\omega_F = \eta(P)\eta(Q), \quad \omega_L = \tfrac{1}{2}(1-\eta(P))(1+\eta(Q)), \quad \omega_U = \tfrac{1}{2}(1-\eta(Q))(1+\eta(P)).$$

They sum to one. Reflection of all angles exchanges P, Q, sends w_i to $-w_i$, and preserves every complete pair and one-body factor. It also preserves either signed constraint. Consequently

$$\mathcal{F}_N^\sigma = F_N + 2L_N. \tag{7.16}$$

The support of L_N has $P \leq 1/2$ and contains only the lower semicircle and short upper endpoint slivers. Lift the latter near π across $-\pi$; the rational pair factors and the constraint are periodic. Its angles then lie in $(-\pi-\delta, \delta)$.

7.3.2 The mixed branch populations

On the support of F_N, $P, Q \geq 1/4$. Since the bump supports stay away from $0, \pi$, the imaginary-part identity gives

$$-\log|Z| \geq c\left(N\varepsilon + \frac{\tau PQ}{N}\right) \geq c/N. \tag{7.17}$$

Every true lower branch has normal displacement $\tau P/N$, and every true upper branch has displacement $\tau Q/N$. The square-root form of γ at $W = 1$ then gives a branch distance at least c/N in the W plane and $|\partial_{\mathsf{S}} \log z_i| \leq C\sqrt{N}$. These statements persist on a common disk $|\mathsf{S} - \mathsf{S}_\varepsilon| \leq c/N^3$: the variation of $\log Z$ is at most $CN^{3/2}c/N^3$, smaller than half the gap in (7.17). Constants can be decreased once to cover all $N \geq 2$.

We estimate the pair product by expanding the angular Schur factors into matchings. Put $A(u, v) = (u - v)/(1 - uv)$. Subtracting the two dispersion equations gives the exact identity $P_{il} = A(z_i, z_l)A(y_i, y_l)$. The scalar Schur–Pfaffian identity expands the product of the $A(y_i, y_l)$ as a sum over matchings when N is even. When N is odd, append the constant ghost entry obtained by setting its variable to -1; its entries are the removable constant one. In each term combine a matched angular factor with its root factor to obtain the complete P_{il}. Every remaining root factor has modulus at most one on the actual contour, since

$$|1 - uv|^2 - |u - v|^2 = (1 - |u|^2)(1 - |v|^2) + 4\Im u \Im v \geq 0 \tag{7.18}$$

in the closed lower half disk. This is the same scalar matching identity as in Lemma 3.1, with no strict contraction required here.

Let B_0 count coordinates in fixed neighborhoods of the two true branches. Each contributes one to $P + Q$, so $B_0 \leq P + Q$. Same-branch complete pairs are bounded: the even inverse in the branch coordinate cancels $\gamma_i + \gamma_l$, leaving $\gamma_i - \gamma_l$. All pairs away from opposite branches are also bounded by compactness. Opposite-branch coordinates lie in a common sector of width less than π, and hence

$$|1 - z_i z_l| \geq c\sqrt{(P + Q)/N}, \qquad |P_{il}| \leq CN/(P + Q) \leq CN/B_0.$$

At most $B_0/2$ matched pairs are exceptional. Since $B_0 \log(N/B_0) \leq N/e$, their total extra cost is C^N. There are $(N - 1)!!$ matchings in the even case and $N!!$ in the odd case. Thus both parities satisfy

$$\left|\prod_{i<l} P_{il}\right| \leq C^N N^{(N+1)/2}. \tag{7.19}$$

This bound holds on the protected complex disk as well. Indeed

$$\partial_{\mathsf{S}} A(u, v) = \frac{u_{\mathsf{S}}(1 - v^2) + v_{\mathsf{S}}(u^2 - 1)}{(1 - uv)^2}.$$

Near the branches, $|1 - uv| \asymp |\gamma_i| + |\gamma_l|$, $|u_{\mathsf{S}}| \leq C/|\gamma_i|$, and $|\gamma_i|, |\gamma_l| \geq cN^{-1/2}$. The displayed derivative is $O(N)$; outside these neighborhoods it is no larger. Each unmatched root entry therefore grows by at most $O(N^{-2})$ on the c/N^3 disk. Their $O(N^2)$-fold product costs only a constant. The canceled matched pairs retain their previous bounds there by the same sector estimates.

The one-body factors have the independent majorant

$$f(\theta) = C\left(1 + \operatorname{dist}(\theta, \{b, -b\})^{-1/2}\right).$$

Near a branch $\theta_b \in \{b, -b\}$, use the imaginary part of $W-1$ when $|\theta-\theta_b| \le C(\varepsilon+|h|)$, and its linear real part outside that region. The small complex disk is absorbed by the branch displacement. In addition, the uniform branch distance c/N gives $|\sinh\gamma_i|^{-1} \le C\sqrt{N}$ for every coordinate throughout the mixed support and its protected disk; outside the true branch neighborhoods the factor is bounded by a constant. Eliminate the fixed Nth angle with the constraint, bound its one-body factor by $C\sqrt{N}$, and use the independent L^1 majorants for the others. This gives, for every $N \ge 2$,

$$\int_{\mathbb{H}_N^\sigma} \omega_F \prod_i |\sinh\gamma_i|^{-1}\, d\mu_N \le C\sqrt{N}\,\|f\|_1^{N-1} \le C^N\sqrt{N}.$$

The estimate remains valid when the harmless cutoff is bounded by one and the remaining independent integrations are enlarged to full circles. The product kernel costs CN by (7.17), and the angular Jacobian costs C^N. Cauchy's inequality on the protected disk and $N! \ge (N/e)^N$ give

$$|\partial_{\mathsf{S}}^j F_N(\mathsf{S}_\varepsilon)| \le C_j^N N^{-N/2+C_j}, \qquad N \ge 2. \tag{7.20}$$

For each fixed $N = 2, 3$, the displaced branches and product gap instead give a bounded analytic integrand on the compact mixed support. Every fixed derivative of these pieces is bounded.

7.3.3 The regular inverse and a cell Gaussian bound

Write $z = e^{-i\varphi}$, $\gamma = i\varphi$, and set $v = \varphi + w$.

Lemma 7.3 (One-sided inverse)**.** *There is a fixed complex neighborhood of the limiting one-sided curve on which $w = U(\mathsf{S}, v)$ is analytic with bounded fixed joint jets. On a fixed convex neighborhood $\mathcal{D}_R$ of the right v-interval, including a short negative branch segment, $\Re U_{vv} \ge c > 0$. All the actual one-sided images, including the endpoint slivers, lie in these neighborhoods when the fixed geometric choices are small.*

Proof. On $[-b, 0]$, $g(\theta) = \arccos(\mathsf{S}_* - \cos\theta)$ has $g' = -\sin\theta/\sin g \ge 0$, and $g'(0) = 0$. Thus $v = g(\theta)+\theta$ is strictly increasing, also on a short upper sliver. At the branch, $w = -\arccos(\mathsf{S}-\cos\varphi)$ is even analytic in φ, and $\mathrm{d}(w+\varphi)/\,\mathrm{d}\varphi = 1$ at zero. The explicit right inverse is

$$U(\mathsf{S}, v) = \frac{v}{2} - \arccos\frac{\mathsf{S}}{2\cos(v/2)}. \tag{7.21}$$

Its limiting real interval is strictly inside $(-2\alpha, 2\alpha)$, because $b < 2\alpha$. With $x = v/2$ and $D = 4\cos^2 x - \mathsf{S}_*^2 > 0$, differentiation gives

$$U_{vv}(\mathsf{S}_*, v) = \frac{\mathsf{S}_*}{4\cos^2 x\sqrt{D}} + \frac{\mathsf{S}_*\sin^2 x}{D^{3/2}} > 0.$$

Compactness preserves a positive real part on a thin convex rectangle. On the left arc $\varphi = -i\gamma$, $\gamma > 0$; the real part of v is θ, and $1 + \varphi_w$ cannot vanish because φ_w is purely imaginary. The left and right embedded arcs meet in the regular branch chart. Their local analytic inverses consequently patch on a sufficiently small neighborhood of the compact arc. Short endpoint extensions preserve injectivity. Square-root continuity near the branch, and ordinary continuity elsewhere, then put every sufficiently small deformed image inside the chosen domains. □

Put $\phi(\mathsf{S},v)=v-U(\mathsf{S},v)$. With the square-root branch fixed by $\sqrt{4\cos^2(v/2)-\mathsf{S}^2}=\sin\phi-\sin U$, implicit differentiation gives

$$U_v=\frac{\sin\phi}{\sqrt{4\cos^2(v/2)-\mathsf{S}^2}},\quad U_{\mathsf{S}}=\frac{1}{\sqrt{4\cos^2(v/2)-\mathsf{S}^2}},\quad a(\mathsf{S},v):=\frac{U_v}{\sinh(i\phi)}=\frac{-i}{\sqrt{4\cos^2(v/2)-\mathsf{S}^2}}. \tag{7.22}$$

The denominator is separated on the fixed inverse neighborhoods. The complete pair is analytic there, including at the double branch: the dispersion identity also writes it locally as

$$P(v_i,v_l)=-\frac{\sin^2((\phi_i-\phi_l)/2)}{\sin^2((U_i+U_l)/2)},$$

whose denominator is nonzero near $U_i=U_l=-b$. Away from that double branch the original pair denominator is separated. The diagonal zero therefore gives the global regular factorization

$$P(v_i,v_l)=(v_i-v_l)^2A(\mathsf{S},v_i,v_l), \tag{7.23}$$

where A and its fixed joint jets are bounded on a smaller neighborhood of the one-sided arc squared.

Lemma 7.4 (Cell contraction and retained collision factors)**.** *On a fixed common complex neighborhood of the one-sided arc, for every fixed ℓ the quotient*

$$\frac{\prod_{i<l\le N}P_{il}}{\prod_{1\le i<l\le\ell}(v_i-v_l)^2}$$

is analytic, and each of its fixed joint jets is bounded by $C_{r,\ell}\exp(-\kappa N^2+C_{r,\ell}N)$ for $N\ge\ell$. The same bound holds after multiplying by $\prod_i a(\mathsf{S},v_i)$. It applies on the enlarged right rectangle with arbitrary actual occupancy fractions in $[0,1]$.

Proof. On the limiting arc $-\pi\le\theta\le 0$, both $y=e^{i\theta}$ and z are in the closed lower half disk: z is on its unit semicircle for $-b\le\theta\le 0$ and on $(0,1]$ on the remaining segment. Equations (7.18) and $P=A(z_i,z_l)A(y_i,y_l)$ give $|P|\le 1$. Apparent endpoint quotients are filled using the analytic complete pair. Since $P(v,v)=0$, finitely many cells, say K, cover the compact regular arc so that same-cell pairs have $|P|\le\eta<1$. Choose a small common complex neighborhood in (S,v): same-cell pairs then have modulus at most $\eta'<1$, and all pairs have modulus at most $M=1+o(1)$. For populations $N_1,\dots,N_K$,

$$\sum_a\binom{N_a}{2}\ge\tfrac12(N^2/K-N),\qquad\prod_{i<l}|P_{il}|\le M^{\binom{N}{2}}(\eta'/M)^{(N^2/K-N)/2}.$$

The neighborhood can be fixed small enough that the coefficient of N^2 in its logarithm is negative. This proves $\exp(-\kappa N^2+CN)$, including on slightly larger fixed polydisks.

Factor the internal ℓ pairs by (7.23). They cost a fixed constant, their cross pairs with the remaining coordinates cost $C^{\ell(N-\ell)}$, and the remaining complete pair product retains its Gaussian bound. Cauchy's inequality on the fixed polydisks gives all asserted joint jets. The bounded one-body product adds only C^N.

The same neighborhood contains the image of the enlarged right rectangle used in coarea. Fix the cells and inverse-domain margins on the undeformed compact arc, then choose the endpoint slivers and negative branch segment inside those margins. In this rectangle every $\rho=P/N$ and $\upsilon=Q/N$ is in $[0,1]$, and $|h_i|\le\tau$. Away from the branch the change in v_i is $O(\varepsilon+\tau)$; near it the uniform square-root bound makes the change $O(\sqrt{\varepsilon+\tau})$. Choosing τ and then ε_0 small puts all such images in the same fixed neighborhood. □

7.3.4 The exact current and its joint jets

For $\chi_{\mathsf{S}}(\theta) = (v_i)$ define the scalar

$$J = \det D_\theta \chi_{\mathsf{S}} = \det M \prod_i U_{v,i}^{-1}.$$

The physical density equals Jf, where

$$f = \frac{1}{N!(2\pi)^{N-1}} \frac{1+\sigma Z}{1-\sigma Z} \prod_i a(\mathsf{S}, v_i) \prod_{i<l} P_{il}. \tag{7.24}$$

This is an equality of scalar coefficients on the original constrained angles: $J\prod a = (\det M)\prod(\sinh\gamma_i)^{-1}$. No intrinsic complex measure on a new constrained v hyperplane is asserted.

For a regular-coordinate field B put

$$V = -M^{-1}(U_{\mathsf{S}} + U_v B). \tag{7.25}$$

If $\sum_i U_{v,i}B_i = -\sum_i U_{\mathsf{S},i}$, then $\sum_i V_i = 0$. If also $\sum_i B_i = 0$, the operator $\partial_{\mathsf{S}} - V\cdot\nabla_\theta$ freezes $\sum_i \varphi_i$. Indeed $\sum\varphi_i = \sum v_i - \sum w_i$, and $\sum w_i = \sum\theta_i$. The exact scalar current identity is

$$\partial_{\mathsf{S}}(Jb) - \mathrm{div}_\theta(VJb) = J\left(E_{\mathsf{S}}b + \sum_i B_i E_i b + \Big(\sum_i E_i B_i\Big)b\right). \tag{7.26}$$

Here b may already be a parameter-dependent transported amplitude. The commuting regular frame is

$$E_i = \sum_l (M^{-1})_{li} U_{v,i}\partial_{\theta_l}, \qquad E_{\mathsf{S}} = \partial_{\mathsf{S}} + \sum_l (M^{-1}U_{\mathsf{S}})_l \partial_{\theta_l}.$$

It satisfies $E_i v_l = \delta_{il}$ and $E_{\mathsf{S}} v_l = 0$. Every commutator is a linear combination of the angular derivatives, annihilates all v_l, and hence vanishes. Thus this frame applies also to smooth real cutoffs without analytically continuing them. To verify (7.26), differentiating the coordinate Jacobian and using Jacobi's determinant formula gives $J\,\mathrm{div}_v B = \partial_{\mathsf{S}}J - \mathrm{div}_\theta(JV)$; the product rule gives the displayed identity. Tangency makes its ambient divergence exactly the constrained one, since

$$\sum_{i<N}(\partial_i - \partial_N)V_i = \sum_i \partial_i V_i.$$

Integration by parts is applied to the real and imaginary parts of these complex currents.

Choose one smooth right cutoff r, equal to one for $\theta + b \geq -\delta_B/2$ and zero for $\theta + b \leq -\delta_B$, with its periodic return in the excluded upper plateau. Decompose $1 = \prod_i r_i + (1 - \prod_i r_i)$ on the one-sided term and telescope the second part. Each of its at most N terms has a named compact left anchor a, also on all cutoff jets. It has $|z_a| \leq q_0 < 1$ and $|U_{v,a}| \geq c$. Use

$$B_a = -\frac{\sum_i U_{\mathsf{S},i}}{U_{v,a}}, \qquad B_i = 0 \ (i \neq a). \tag{7.27}$$

This gives tangency; phase freezing is unnecessary because $|1-\sigma Z| \geq 1-q_0$. All fixed regular-frame jets of B, V and the cutoffs have polynomial N bounds. The cell Gaussian, the bounded regular one-body factors, and (7.26) then bound every fixed derivative of this whole piece by

$$\frac{C_j N^{C_j}}{N!}\exp(-\kappa N^2 + C_j N). \tag{7.28}$$

In fact, eliminate the compact anchor angle with the constraint. Its Jacobian factor is bounded, and every remaining factor has the independent integrable majorant $C(1+\operatorname{dist}(\theta,-b)^{-1/2})$. Their integrals cost only C^N. The product kernel and its transported derivatives have polynomial N bounds on this anchored support. This argument applies also at each fixed $N=2,3$.

We now work on the full right part, with all angles in $I=[-b-\delta_B,\delta]$. Set $u_i=\theta_i+b$ and, on a diameter shell $r_0\le d:=\max\theta_i-\min\theta_i\le 2r_0$, put

$$\mu=N^{-1}\sum_i u_i,\qquad T=|\mu|+r_0+\varepsilon,\qquad \Delta=r_0/\sqrt{T}. \tag{7.29}$$

These are pointwise envelopes; no derivative of a mean, diameter, or scale is ever taken.

The branch normal form, for a common outward displacement $h\ge 0$, is

$$\begin{gathered}1-W=\sin b\,u-i\beta+R,\qquad c(\varepsilon+h)\le\beta\le C(\varepsilon+h),\\ R\in\mathbb{R},\qquad |R|\le C(u^2+\varepsilon^2+h^2),\qquad \varphi=\sqrt{2(1-W)}(1+O(1-W)).\end{gathered} \tag{7.30}$$

Indeed $W=\mathsf{S}-\cosh(h+i(-b+u))$ and $\beta=\Im\mathsf{S}+\sinh h\sin(b-u)$; its real part has the displayed Taylor expansion. The physical square root lies in a small enlargement of the fourth quadrant. In particular

$$|1+\varphi_w(w_i)|\le C(|u_i|+\varepsilon)^{-1/2},\quad -\Im\varphi_i\ge c\sqrt{|u_i|+\varepsilon}\quad (u_i\le 0),$$
$$\int_{\mu-2r_0}^{\mu+2r_0}(|u|+\varepsilon)^{-1/2}\,\mathrm{d}u\le Cr_0/\sqrt{T}.$$

The interval bound follows by comparability if $|\mu|\ge 4r_0$; otherwise enlarge to $[-6r_0,6r_0]$ and rationalize $\sqrt{6r_0+\varepsilon}-\sqrt{\varepsilon}$. Joining two angles along their one-variable curve with the actual $\rho=P/N,\upsilon=Q/N$ held fixed gives

$$|v_i-v_l|\le C\Delta,\qquad |U_{v,i}|\le C\sqrt{T}. \tag{7.31}$$

The joining derivative has the additional bounded factor $1-i\tau(\rho m'-\upsilon p')$. Near the lower branch its outward shift is the common $\tau\rho$, so the preceding interval bound applies; elsewhere it is regular. For the second bound, if $P=0$ then every actual $h_i=0$, and the branch form and (7.22) apply directly. If $P>0$, some actual angle is positive, so $\max u_i\ge b$ and $T\ge b/2$; boundedness of U_v proves the same result. These arguments allow all actual occupancy fractions in $[0,1]$.

Use the filled analytic divided differences

$$\mathcal{U}_{il}=\int_0^1 U_v(v_l+t(v_i-v_l))\,dt,\qquad \mathcal{E}_{il}=\int_0^1 U_{vv}(v_l+t(v_i-v_l))\,dt.$$

The convex inverse domain gives $\Re\mathcal{E}_{il}\ge c$. Define the analytic field

$$\begin{gathered}D_U=\sum_{i<l}(U_i-U_l)^2,\qquad H_{il}=\frac{(v_i-v_l)\mathcal{U}_{il}^2}{\mathcal{E}_{il}},\qquad G_i=\sum_l H_{il},\\ A=-\sum_i U_{\mathsf{S},i},\qquad B_i=AG_i/D_U.\end{gathered} \tag{7.32}$$

There is no pair pole: H_{il} is analytic also at $v_i=v_l$. Away from equality it equals $(U_i-U_l)^2/(U_{v,i}-U_{v,l})$. Antisymmetry and pairing give

$$\sum_i G_i=0,\qquad \sum_i U_{v,i}G_i=D_U,\qquad \sum_i B_i=0,\qquad \sum_i U_{v,i}B_i=-\sum_i U_{\mathsf{S},i}.$$

Thus the field has both exact transport identities.

Lemma 7.5 (Protected analytic variance and joint jets)**.** *On each actual right shell,*

$$\Re D_U \geq c\sum_{i<l}(\theta_i-\theta_l)^2 \geq cT\Delta^2 > 0. \tag{7.33}$$

For every fixed total derivative order s,

$$|\mathcal{D}^s B_i| + |\mathcal{D}^s V_i| \leq C_s N^{C_s}\Delta^{-1-s}. \tag{7.34}$$

The field is regular at every partial diagonal.

Proof. On the actual contour $U_i = w_i = \theta_i - ih_i$, and the real balanced shifts satisfy $|h_i - h_l| \leq C\tau|\theta_i-\theta_l|$ at fixed actual occupancies. Hence $\Re(U_i-U_l)^2 = (\theta_i-\theta_l)^2-(h_i-h_l)^2 \geq (1-C^2\tau^2)(\theta_i-\theta_l)^2$. Small fixed τ proves the denominator bound.

Freeze the actual values of T, Δ and consider the polydisk

$$|\delta\mathsf{S}|,\ |\delta v_i| \leq c_0\Delta/N^2.$$

Choose c_0 small so it lies in the fixed inverse neighborhoods. Throughout this disk, $|v_i - v_l| \leq C\Delta$, $|U_v| \leq C\sqrt{T}$, and $|U_{\mathsf{S}}(v_i) - U_{\mathsf{S}}(v_l)| \leq C\Delta$. Indeed the last bound is an analytic divided difference; the middle one follows from its actual endpoint bound and bounded joint jets, since $\Delta \leq \sqrt{T}$. Along the straight parameter and coordinate segments, the common parameter change therefore moves each $U_i - U_l$ by $O(\Delta|\delta\mathsf{S}|)$, and the two coordinate changes move it by $O(\sqrt{T}(|\delta v_i| + |\delta v_l|))$. Their total change is at most Cc_0r_0/N^2. Since each original difference is $O(r_0)$, summing the $N(N-1)/2$ squared differences gives

$$|D_U^{\mathrm{new}} - D_U| \leq Cc_0r_0^2 \leq \tfrac{1}{2}\Re D_U.$$

Thus $|D_U^{\mathrm{new}}| \geq cr_0^2$ on this whole polydisk. Also $|\mathcal{U}_{il}| \leq C\sqrt{T}$ and $|\mathcal{E}_{il}| \geq c$ there, so $|H_{il}| \leq Cr_0\sqrt{T} = CT\Delta$, $|A| \leq CN$, and $|B_i| \leq CN^2/\Delta$. Cauchy's inequality now gives all fixed joint jets at once:

$$|\partial_{\mathsf{S}}^a\partial_v^\beta B_i| \leq C_s N^{2+2s}\Delta^{-1-s}, \qquad s = a + |\beta|. \tag{7.35}$$

The commuting-frame coefficients $M^{-1}\operatorname{diag} U_v$ and $M^{-1}U_{\mathsf{S}}$ have polynomially bounded fixed joint jets: the real angular jets of M and analytic jets of U are bounded, and differentiating M^{-1} inductively bounds its jets. The frame formulas use U_v with a nonnegative power. The same induction bounds the fixed smooth cutoff jets. Formula (7.25), the product rule, and (7.35) give the same field bound for V. The denominator is separated off full equality, including at partial diagonals where the analytic quotients H_{il} extend. □

Fix now $\ell = n_* + 1$ and $D_\ell = \ell(\ell-1)$. Factor the first ℓ collision variables using Lemma 7.4. For the initial smooth weighted regular amplitude b_0 after removing the product kernel,

$$|\mathcal{D}^s b_0| \leq \frac{C_s N^{C_s}}{N!} e^{-\kappa N^2 + C_s N}\Delta^{D_\ell - s}, \qquad s \leq 2k.$$

The same statement includes parameter derivatives, which preserve the explicit collision polynomial. Iterate the exact current with

$$b_{j+1} = \mathcal{T}b_j, \qquad \mathcal{T}b = E_{\mathsf{S}}b + \sum_i B_iE_ib + \Big(\sum_i E_iB_i\Big)b. \tag{7.36}$$

This recurrence differentiates every parameter-dependent descendant coefficient and cutoff. Because the field freezes $\sum \varphi_i$, the product kernel is unchanged. Induction using (7.34) gives

$$|\mathcal{D}^s b_j| \le \frac{C_j N^{C_j}}{N!} e^{-\kappa N^2 + C_j N} \Delta^{D_\ell - 2j - s}, \qquad s \le 2(k-j). \tag{7.37}$$

Indeed E_{S} loses at most one scale power, while $B_i E_i$ and $(E_i B_i)$ each lose at most two. Leibniz' rule gives the same count for joint jets. Only polynomially many index choices occur at each fixed order.

The boundary flux vanishes. At fixed $N, \varepsilon > 0$ puncture the finitely many full-equality points by angular balls of radius r. All geometric Jacobians and the product kernel are then smooth; $V = O(r^{-1})$, and the retained collision degree is D_ℓ. The flux after j transports is bounded by a constant times $r^{D_\ell + N - 3 - 2j}$, which tends to zero for every $N \ge n_* + 2$ and $j \le k$. Partial diagonals require no puncture by (7.33). The original smooth weights vanish at the outer chart boundaries; no diameter cutoff has been differentiated. Thus the jth derivative of the complete right term is the integral of its transported density, bounded by

$$\frac{C_j N^{C_j}}{N!} e^{-\kappa N^2 + C_j N} \Delta^{D_\ell - 2j} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - \sigma Z|}. \tag{7.38}$$

7.3.5 Coarea with coupled occupancies

The phase derivative in the coarea estimate includes the derivatives of both occupancies.

First, for $G = x - ia$, $a \ge a_0 > 0$, and fixed $|\vartheta| < \pi/2$, put $\nu = \Re(e^{-i\vartheta} G)$ and $P_\vartheta = 2\pi \cos\vartheta$. If $a_0 \le 1$, then

$$\frac{1}{|1 - e^{-iG}|} \le \frac{C_\vartheta}{a_0 + \operatorname{dist}(\nu, P_\vartheta \mathbb{Z})}. \tag{7.39}$$

For $a \le 1$, this follows from $|1 - e^{-a-ix}| \ge c(a + \operatorname{dist}(x, 2\pi\mathbb{Z}))$ and $|\nu - 2\pi n \cos\vartheta| \le \cos\vartheta |x - 2\pi n| + |\sin\vartheta| a$. For $a \ge 1$, the left denominator is separated and the proposed right denominator is bounded above. Integrating the majorant over an interval of length L costs at most $C_\vartheta(1 + L)(1 + |\log a_0|)$. For $\sigma = -1$ replace G by $G - \pi$. On the actual contours $-\log|Z| \ge cN\varepsilon$: write $\varphi_i = x_i - i a_i$, and use $\Im W_i = \sin x_i \sinh a_i \le C a_i$ on the bounded phase range. Apply the lemma with $a_0 = \min(1, cN\varepsilon)$. Its logarithmic cost is at most $C(1 + L)(1 + |\log\varepsilon|)$.

Lemma 7.6 (Complete right-shell coarea)**.** *On each fixed-mean lift of the full right rectangle $I^N \cap \mathbb{H}_N^\sigma$, with the actual balanced occupancies, the integral over any diameter shell in* (7.29) *obeys*

$$\int_{\substack{\sum u_i = N\mu \\ r_0 \le d \le 2r_0}} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - \sigma Z|} \prod_{i<N} \mathrm{d}\theta_i \le C^N N^C \Delta^{N-3} (1 + |\log\varepsilon|). \tag{7.40}$$

There are at most CN constrained means. The constants are uniform in each of them; their subsequent sum costs only this polynomial factor.

Proof. Write $m_0 = \min u_i$, $M_0 = \max u_i$, and first suppose $0 \le m_0 < M_0$. With other angles fixed, the constraint fixes $m_0 + M_0$. Put $G = \sum_i \varphi_i$ and $\vartheta = 3\pi/8$. We claim

$$\Re(e^{-i\vartheta}\, \mathrm{d}G/\,\mathrm{d}m_0) > 0, \qquad \frac{|(1 + \varphi_w(w_{m_0}))(1 + \varphi_w(w_{M_0}))|}{\Re(e^{-i\vartheta}\, \mathrm{d}G/\,\mathrm{d}m_0)} \le C \frac{\sqrt{M_0 + \varepsilon}}{M_0 - m_0}. \tag{7.41}$$

For the one-variable comparison curve, fix the values $\rho = P/N, \upsilon = Q/N$ at the configuration and set $f(u) = -b + u - i\tau\{\rho m(-b+u) - \upsilon p(-b+u)\}$, $E = \varepsilon + \tau\rho$. For $u \geq 0$, the branch normal form and compactness on the rest of I give

$$|\varphi_w(f(u))| \leq C(u+E)^{-1/2}, \qquad |\varphi_{ww}(f(u))| \asymp (u+E)^{-3/2}.$$

The inverse supplies $-\varphi_{ww} = U_{vv}/U_v^3$. For $u \geq 0$ the argument of U_v is in a small enlargement of $[-\pi/4, 0]$, while U_{vv} has small argument. Hence $-\varphi_{ww}$ has uniformly positive $3\pi/8$ projection. If $a = f' = 1 + O(\tau)$ and $B(u) = \varphi_w(f(u))a(u)$, then $-B' = -\varphi_{ww}(f)a^2 - \varphi_w(f)a'$ retains that projection for small τ. Consequently

$$\begin{gathered} \Re\{e^{-i\vartheta}[B(m_0) - B(M_0)]\} \geq cL, \\ L = \int_{m_0}^{M_0} (u+E)^{-3/2}\,\mathrm{d}u = \frac{2(M_0 - m_0)}{\sqrt{m_0+E}\sqrt{M_0+E}(\sqrt{m_0+E} + \sqrt{M_0+E})}. \end{gathered} \tag{7.42}$$

Now restore all occupancy derivatives. Put $B_m^{\mathrm{occ}} = \sum_l m(\theta_l)\varphi_w(w_l)$ and $B_p^{\mathrm{occ}} = \sum_l p(\theta_l)\varphi_w(w_l)$. Direct differentiation of (7.15) gives the exact identity

$$\partial_{\theta_i} G = \varphi_w(w_i)[1 - i\tau(\rho m_i' - \upsilon p_i')] - \frac{i\tau p_i'}{N} B_m^{\mathrm{occ}} + \frac{i\tau m_i'}{N} B_p^{\mathrm{occ}}. \tag{7.43}$$

Every actual $u_l \geq m_0$, so $(|B_m^{\mathrm{occ}}| + |B_p^{\mathrm{occ}}|)/N \leq C(m_0+E)^{-1/2}$. Taking $\partial_{\theta_{\min}} - \partial_{\theta_{\max}}$, both bump-derivative differences are $O(M_0 - m_0)$. The entire occupancy correction is therefore at most $C\tau(M_0 - m_0)(m_0+E)^{-1/2} \leq C\tau L$, by the exact expression for L and the fixed upper bound on $M_0 + E$. Small fixed τ absorbs it into the positive projection. The numerator of (7.41) is at most $C[(m_0+E)(M_0+E)]^{-1/2}$, giving ratio $C\sqrt{M_0+E}/(M_0 - m_0)$. If $P = 0$, $E = \varepsilon$; if $P > 0$, an actual positive angle gives $M_0 \geq b$, and hence $E \leq C(M_0 + \varepsilon)$. This proves (7.41) with all actual occupancies, including on the enlarged rectangle.

After transport, bound every smooth coefficient by its proved envelope and enlarge to this full rectangle. All its occupancy fractions lie in $[0,1]$, the balanced displacement preserves $\Im W_i \geq \Im \mathsf{S}$, and Lemma 7.4 has already placed its images in the regular domains. Fix a constrained mean $\mu = b + (2\pi l + a_\sigma)/N$ and the two extreme indices; at most CN^3 such choices occur. Eliminate the maximum coordinate with the constraint. With the remaining $N-2$ coordinates fixed, conditions specifying the extrema, the shell, and a nonnegative minimum are linear interval conditions in m_0, with $M_0 = A - m_0$. Thus the fiber is one interval, and its projected phase is injective by (7.41). Its phase range is at most CN. Equations (7.39) and (7.41) bound this extreme-coordinate integral by

$$CN\frac{\sqrt{T}}{r_0}(1 + |\log\varepsilon|).$$

If $m_0 < 0$, its branch attenuation from (7.30) instead gives the direct bound

$$\frac{|(1+\varphi_w(w_{m_0}))(1+\varphi_w(w_{M_0}))|}{|1 - \sigma Z|} \leq \frac{C}{(|m_0| + \varepsilon)\sqrt{|M_0| + \varepsilon}}.$$

With $a = |m_0| + \varepsilon$ and $c = |M_0| + \varepsilon$, $a + c \geq r_0$, and

$$\frac{1}{a\sqrt{c}} \leq \frac{1}{\sqrt{r_0}}\left(\frac{1}{a} + \frac{1}{c}\right).$$

Each reciprocal integrates to $C(1 + |\log\varepsilon|)$ on the bounded slice $M_0 = A - m_0$. Since $T \geq r_0$, this gives the same extreme pair envelope $C\sqrt{T}/r_0(1 + |\log\varepsilon|)$. Only now replace the other $N-2$

Jacobian factors by the independent majorants and integrate each over $[\mu - 2r_0, \mu + 2r_0]$. Their total cost is $(Cr_0/\sqrt{T})^{N-2}$. Combining the extreme pair, the ordinary factors, and the polynomial number of cells proves (7.40). □

7.3.6 One sum for all higher particle orders

Combining (7.38) with Lemma 7.6 bounds a right shell at a fixed mean by

$$\frac{C_j N^{C_j}}{N!} e^{-\kappa N^2 + C_j N} \Delta^{D_\ell + N - 3 - 2j} (1 + |\log \varepsilon|). \tag{7.44}$$

Its exponent is positive for every $N \geq n_* + 2$ and $j \leq k$: it is at least $n_* + N - 2$ in the odd case and $n_* + N - 1$ in the even case. For fixed μ, ε,

$$\frac{\Delta(r_0/2)}{\Delta(r_0)} = \frac{1}{2}\sqrt{\frac{|\mu| + r_0 + \varepsilon}{|\mu| + r_0/2 + \varepsilon}} \in [1/2, 1/\sqrt{2}].$$

The dyadic sum is therefore bounded uniformly in the mean by a geometric series. Its largest possible scale is fixed; its power costs only C^N times a fixed constant. This is absorbed into the linear exponential in (7.44). Summing the at most CN means adds only a polynomial factor. The mixed, anchored, and right pieces consequently obey the same bound for all higher orders,

$$|\partial_{\mathsf{S}}^j \mathcal{F}_N^\sigma(\mathsf{S}_\varepsilon)| \leq C_j (1 + |\log \varepsilon|) \left(C_j^N N^{-N/2 + C_j} + \frac{N^{C_j}}{N!} e^{-\kappa N^2 + C_j N} \right). \tag{7.45}$$

Both envelopes are summable. The exact split (7.16) proves Theorem 7.2. The parameter-change assertion follows from the finite chain rule.

Proposition 7.7 (Transport estimates at fixed particle order)**.** *For each fixed $N \geq 2$ and every fixed derivative order, the mixed piece is bounded and extends holomorphically across S_* on its deformed contour. A smooth one-sided term with a compact left anchor has bounded derivatives of every fixed order. A smooth right-supported term whose angular diameter is bounded below by a fixed $d_0 > 0$ has derivatives $O(1 + |\log \varepsilon|)$ of every fixed order. The assertions allow arbitrary fixed smooth angular cutoffs with those supports.*

Proof. For the fixed mixed order, both true branches and the product denominator have uniform gaps on the compact support, as proved above; the same small parameter disk works for the whole integrand. This includes $N = 2, 3$ directly. The anchored field (7.27) and its integrable independent Jacobian majorants prove the second assertion at any fixed derivative budget. On a right support with $d \geq d_0$, all jets of the covariance field are bounded and there is no collision puncture. Iterate (7.36) with that budget, and then apply Lemma 7.6; the finitely many dyadic scales bounded away from zero cost $O(1+|\log \varepsilon|)$. For $N = 2$ the coarea keeps both extreme factors and has zero remaining one-body integrations, so the same argument applies. All differentiated cutoff factors are included in the exact recurrence. □

7.4 All lower coefficients and the complete finite complement

Lemma 7.8 (Finite complement at every fixed derivative order)**.** *At any of the signed positive-geometry points just specified, every fixed S derivative of every smaller particle coefficient of the relevant parity is bounded along* (7.11)*. At $n = n_*$ choose a fixed smooth local shape cutoff at*

$\theta_1 = \cdots = \theta_n = -\alpha_+$, *and let* I_n^σ *be the resulting local integral. Then every fixed derivative of* $\mathcal{F}_n^\sigma - 2I_n^\sigma$ *is bounded on the same path. These conclusions also hold for the exterior-coordinate derivatives.*

Proof. We first prove $O(1 + |\log\varepsilon|)$ at every fixed derivative order; constants may depend on this finite particle order and derivative order. The case $n = 1$ follows from (7.7). For all $n \geq 2$ use the whole-torus balanced deformation and exact reflection decomposition from the preceding transport argument:

$$\mathcal{F}_n^\sigma = F_n^\sigma + 2L_n^\sigma.$$

The mixed-population piece has both occupancies at least $1/4$. At this fixed n, both true branches are displaced by at least $\tau/(4n)$, and the product kernel has a fixed damping gap. The complete density is therefore holomorphic and bounded on a fixed parameter neighborhood. This includes $n = 2, 3$ by the same compactness argument.

Each left-root term in L_n^σ has a named compact anchor with $|\rho_a| \leq q_0 < 1$ and $|U_{v,a}| \geq c > 0$. The anchor field $B_a = -\sum_i U_{\mathsf{S},i}/U_{v,a}$, $B_i = 0$ for $i \neq a$, makes the angular representative tangent to the constraint. The current identity (7.26) leaves all fixed regular derivatives bounded and the product kernel separated. Eliminate the compact anchor angle using the exact constraint. Its Jacobian factor is bounded; every remaining factor is bounded by $C(1 + \operatorname{dist}(\theta, -b)^{-1/2})$, independently integrable. This proves boundedness of every fixed derivative of the entire left-root part, including its cutoff derivatives.

It remains to analyze the right part. Choose its branch width below one quarter of the finite mean gap for all relevant $n \leq n_*$. Every constrained equality configuration on this support is then a compact regular point away from the branch. At full equality the balanced shift is exactly zero: writing its upper and lower bumps as p, m, one has $P = np(\theta_0)$, $Q = nm(\theta_0)$ and $Pm(\theta_0) - Qp(\theta_0) = 0$. Its product denominator can vanish only when

$$\mathsf{S}_* = \cos\theta_0 + \cos\varphi_0, \qquad n\theta_0, n\varphi_0 \in 2\pi\mathbb{Z} + a_\sigma.$$

The prime arithmetic excludes every smaller order. At n_* it leaves only $-\alpha_+$ on the one-sided support; $+\alpha_+$ is in the excluded upper plateau. Each noncritical equality admits a fixed neighborhood on which all roots and pair factors are regular and $1 - \sigma Z$ is separated. Its localized integral is holomorphic through the parameter point.

Remove these finitely many neighborhoods and, at n_*, the critical chart. Take the latter wholly inside the lower plateau $m = 1, p = 0$, where the deformation is identically zero, the one-sided weight is one and the right cutoff is one. The remaining right support and all its cutoff jets have angular diameter $d \geq d_0 > 0$. The collective field in Proposition 7.7 has bounded fixed jets there, with no collision puncture or flux. After any fixed number of applications of (7.26), the density is bounded by

$$C_{n,j} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - \sigma Z|}.$$

Apply Lemma 7.6 to the two extreme-coordinate Jacobian factors and bound the remaining factors independently. There are finitely many means and diameter shells with $d \geq d_0$, so their integrals are $O(1 + |\log\varepsilon|)$. When $n = 2$ there are no remaining one-body integrals, and the same two-extreme estimate applies. Smooth cutoff fibers are enlarged to the geometric coarea intervals after differentiation.

For the strengthening from logarithmic to bounded derivatives, let R be any complete lower coefficient or the complete remainder $\mathcal{F}_{n_*}^\sigma - 2I_{n_*}^\sigma$. The contours and weights defining R are fixed

independently of S, so it is holomorphic on the one-sided sector. For each fixed j,

$$\partial_{\mathsf{S}}^{j} R(\mathsf{S}_{\varepsilon}) = \partial_{\mathsf{S}}^{j} R(\mathsf{S}_{\varepsilon_1}) - \int_{\varepsilon}^{\varepsilon_1} \mathsf{S}'_u \, \partial_{\mathsf{S}}^{j+1} R(\mathsf{S}_u)\, du.$$

The logarithmic bound on the next derivative is integrable and S'_u is bounded, so each fixed derivative of R has a finite radial limit. The finite chain rule gives exterior-coordinate derivatives. The factor two is exactly the reflection factor in $\mathcal{F}_n^{\sigma} = F_n^{\sigma} + 2L_n^{\sigma}$. □

7.5 The first singular coefficient with its exact phase

The following calculation uses only the complete product (7.1). Let $N = n_*$, $\alpha = \alpha_+$ and $N\alpha \in 2\pi\mathbb{Z} + a_\sigma$. In its lower chart set

$$\theta_i = -\alpha + u_i, \qquad u_N = -\sum_{i<N} u_i, \qquad Q(u) = \sum_i u_i^2, \qquad \mathcal{V}_N(u) = \prod_{i<j} (u_i - u_j).$$

This is the literal angular constraint, with coordinate measure $\prod_{i<N} du_i/(2\pi)$. In this chart the physical root is $\rho_i = e^{-\mathrm{i}\varphi_i}$, where $\varphi_i = \arccos(\mathsf{S} - \cos\theta_i)$. At the central point its derivatives are

$$\varphi = \alpha, \qquad \varphi_{\mathsf{S}} = -\frac{1}{\sin\alpha}, \qquad \varphi_\theta = 1, \qquad \varphi_{\theta\theta} = -2\cot\alpha.$$

Writing $h = \mathsf{S} - \mathsf{S}_*$ and $d_\alpha = \cot\alpha > 0$, the zero-sum constraint cancels both the linear shape term and its mixed parameter derivative, so

$$\sum_i \varphi_i = N\alpha - \frac{N}{\sin\alpha} h - d_\alpha Q(u) + O\left(|h|^2 + |h|Q(u) + \sum_i |u_i|^3\right).$$

Consequently, for $q_\sigma = 1 - \sigma Z$,

$$q_\sigma(\mathsf{S}_\varepsilon, u) = 2N\varepsilon - \mathrm{i} d_\alpha Q(u) + O\left(\varepsilon^2 + \varepsilon Q(u) + Q(u)^2 + \sum_i |u_i|^3\right), \tag{7.46}$$

$$\begin{aligned} \partial_{\mathsf{S}} q_\sigma(\mathsf{S}_*, 0) &= -\frac{\mathrm{i}N}{\sin\alpha} =: D_{\mathsf{S}}, \\ \partial_t q_\sigma(\zeta, 0) &= 2N\zeta^{-1} =: D_t. \end{aligned} \tag{7.47}$$

All constants here may depend on the fixed N. The two leading terms in (7.46) have perpendicular phases. By choosing a small fixed real shape ball and then ε_0, the remainder is less than half their lower bound. Thus

$$|q_\sigma(\mathsf{S}_\varepsilon, u)| \geq c(\varepsilon + Q(u)). \tag{7.48}$$

For the fixed shape cutoff χ, the chart integral is exactly

$$I_N^{\sigma}(\mathsf{S}) = \int_{\mathbb{R}^{N-1}} \frac{\chi(u)\mathcal{V}_N(u)^2 A_N(\mathsf{S}, u)}{q_\sigma(\mathsf{S}, u)}\, du_1 \cdots du_{N-1}, \tag{7.49}$$

with analytic amplitude and

$$A_N(\mathsf{S}_*, 0) = \mathrm{i}^{-N^2} \frac{2^{1-N(N-1)}}{N!(2\pi)^{N-1}\sin^{N^2}\alpha} =: \mathrm{i}^{-N^2} a_N(\alpha), \qquad a_N(\alpha) > 0. \tag{7.50}$$

Indeed every sine numerator gives half an angle difference, every pair denominator tends to $\mathrm{i}\sin\alpha$, every one-body factor to $(\mathrm{i}\sin\alpha)^{-1}$, and $1 + \sigma Z \to 2$. This includes all factorial and Haar normalizations. At fixed shape coordinates every parameter derivative retains the whole squared Vandermonde. The reflected upper chart is exactly equal under angle reflection.

Lemma 7.9 (The nonzero scalar period)**.** *Put*

$$a=\frac{N^2-1}{2},\qquad k=\lfloor a\rfloor,\qquad \nu=k+1-a\in\{\tfrac12,1\},\qquad \mathcal{S}_N=\frac{1}{\sqrt{N}}\int_{\substack{\sum_i\omega_i=0\\ \sum_i\omega_i^2=1}}\mathcal{V}_N(\omega)^2\,d\sigma(\omega)>0.$$

Then

$$\begin{aligned}J_{N,\alpha}&:=\int_{\mathbb{R}^{N-1}}\frac{\mathcal{V}_N(v)^2}{(2N-\mathrm{i}d_\alpha Q(v))^{k+1}}\,dv_1\cdots dv_{N-1}\\&=\frac{\mathcal{S}_N}{2}(2N)^{-\nu}(-\mathrm{i}d_\alpha)^{-a}\,\mathrm{B}(a,\nu)\neq0.\end{aligned}\tag{7.51}$$

The branch in the last line has $\arg(-\mathrm{i}d_\alpha)=-\pi/2$*. For odd prime* $N=p$*,* $k=(p^2-1)/2\equiv0\pmod4$ *and this simplifies to*

$$J_{p,\alpha}=\frac{\mathcal{S}_p}{4pk\,d_\alpha^k}>0.\tag{7.52}$$

Proof. The map $(v_1,\dots,v_{N-1})\mapsto(v_1,\dots,v_N)$, $v_N=-\sum_{i<N}v_i$, has Gram determinant N. Its coordinate measure is therefore Euclidean measure divided by $\sqrt{N}$, exactly as in the constrained angular normalization. Polar coordinates on that hyperplane give the radial power $r^{N^2-2}=r^{2a-1}$. With $x=r^2$,

$$J_{N,\alpha}=\frac{\mathcal{S}_N}{2}\int_0^\infty\frac{x^{a-1}}{(2N-\mathrm{i}d_\alpha x)^{k+1}}\,dx.$$

For a positive real coefficient in place of $-\mathrm{i}d_\alpha$, a real substitution gives the beta integral in (7.51). Continue that coefficient through the fourth quadrant. Throughout the closed quadrant at fixed positive modulus, $|2N+cx|\geq c_0(1+x)$, so dominated convergence extends the same identity to the displayed imaginary coefficient. The radial tail is r^{-2} for even N and r^{-3} for odd N. The angular constant is positive because the zero-sum sphere has an open set of vectors with distinct coordinates.

For odd $N=p$, an antiderivative gives the same period: with $Q_0=2p$ and $c=-\mathrm{i}d_\alpha$,

$$\frac{d}{dx}\left(\frac{cx}{Q_0+cx}\right)^k=kQ_0c^k\frac{x^{k-1}}{(Q_0+cx)^{k+1}}.$$

The bracket has endpoint values zero and one; and $c^k=d_\alpha^k$ because $k\equiv0\pmod4$. This proves (7.52) directly. □

Theorem 7.10 (The complete first coefficient)**.** *For either parity and its signed positive geometry,*

$$\partial_{\mathsf{S}}^k\mathcal{F}_N^\sigma(\mathsf{S}_\varepsilon)=\Lambda_{N,\alpha}^{\mathsf{S}}\varepsilon^{-\nu}+o(\varepsilon^{-\nu}),\tag{7.53}$$

$$(C_N^\sigma)^{(k)}(t_\varepsilon)=\Lambda_{N,\alpha}^{t}\varepsilon^{-\nu}+o(\varepsilon^{-\nu}),\tag{7.54}$$

$$\begin{aligned}\Lambda_{N,\alpha}^{\mathsf{S}}&=2\mathrm{i}^{-N^2}a_N(\alpha)(-1)^kk!D_{\mathsf{S}}^kJ_{N,\alpha}\neq0,\\\Lambda_{N,\alpha}^{t}&=2\mathrm{i}^{-N^2}a_N(\alpha)(-1)^kk!(2N\zeta^{-1})^kJ_{N,\alpha}\neq0.\end{aligned}\tag{7.55}$$

All derivatives below k *are bounded for even* N*. For odd* N*, they are bounded through* $k-2$ *and at most logarithmic at* $k-1$*.*

Proof. Lemma 7.8 disposes of the complete complement. Differentiate (7.49) k times at fixed real shape coordinates. The unique highest pole is

$$\int \frac{\chi(u)\mathcal{V}_N(u)^2(-1)^k k!(\partial_{\mathsf{S}} q_\sigma)^k A_N}{q_\sigma^{k+1}}\, du.$$

Under $u = \sqrt{\varepsilon}\, v$ its total scale is $\varepsilon^{(N^2-1)/2-(k+1)} = \varepsilon^{-\nu}$. Equation (7.48) gives the integrable majorant $C\mathcal{V}_N(v)^2/(1+Q(v))^{k+1}$. Dominated convergence and (7.50) therefore give the single-chart coefficient $\mathrm{i}^{-N^2} a_N (-1)^k k! D_{\mathsf{S}}^k J_{N,\alpha}$. The reflected chart doubles it. This reflection factor is distinct from the factor 2 already in a_N from $1+\sigma Z$.

Every other term has denominator power at most k. Its radial absolute bound is $C\int_0^\delta r^{2a-1}(\varepsilon + r^2)^{-m-1}\, dr$ with $m < k$. It is bounded in the even case, and at worst $O(|\log\varepsilon|)$ in the odd case; both are $o(\varepsilon^{-\nu})$. The same radial powers give the assertions about lower derivatives. Finally $\mathsf{S}'(\zeta) = 2\mathrm{i}\sin\alpha\,\zeta^{-1} \neq 0$. The highest term in the finite chain rule is $(\mathsf{S}')^k \partial_{\mathsf{S}}^k$; all remaining terms are bounded or logarithmic. The identity $D_{\mathsf{S}}\mathsf{S}'(\zeta) = 2N\zeta^{-1}$ proves the stated exterior-coordinate constant. □

In particular the odd coefficient has

$$\Lambda_{p,\alpha}^t = -2\mathrm{i}\, a_p(\alpha) k! (2p)^k \zeta^{-k} J_{p,\alpha}.$$

For an original second-quadrant HT point, apply (7.10). Since $k \equiv 0 \pmod 4$, $\sin\widehat{\alpha} = \sin\alpha$ and $\widehat{\zeta}^{-k} = \overline{\zeta}^{-k}$, its constant has exactly the same formula with $J_{p,\alpha} = \mathcal{S}_p/(4pk|\cot\alpha|^k) > 0$. Thus both HT signs yield nonzero coefficients with the physical ordinary constraint after the exact algebraic transfer.

7.6 Noncancellation for the infinite physical sums

Theorem 7.11 (Isotropic natural boundary)**.** *The full LT normalized response has the unit circle as a natural boundary in its exterior variable s. The full HT normalized response has the unit circle as a natural boundary in its disk variable s. These assertions concern the exact physical sums* (7.3)–(7.4)*.*

Proof. At a fixed prime point, normal convergence permits termwise differentiation. All smaller coefficients have bounded kth derivatives by Lemma 7.8. The first one has the nonzero term in Theorem 7.10, whereas Theorem 7.2 gives

$$\sum_{\substack{N>n_* \\ N \text{ of the chosen parity}}} |\partial_{\mathsf{S}}^j \mathcal{F}_N^\sigma(\mathsf{S}_\varepsilon)| = O(1+|\log\varepsilon|), \qquad 0 \le j \le k.$$

The finite chain rule gives the same tail bound for the exterior derivatives. Since $\varepsilon^\nu(1+|\log\varepsilon|) \to 0$, the entire sum has precisely the nonzero leading singularity $\Lambda^t\varepsilon^{-\nu}$ in its kth derivative. For negative HT dispersion use (7.10) for every coefficient and every required derivative, including the entire tail. This proves the same obstruction for the physical ordinary-sign series there.

A holomorphic continuation through such a point would bound every fixed derivative on a short radial segment, a contradiction. The physical multipliers $(1-s^{-4})^{1/4}$ and $Q_H(t)$ are locally analytic and nonzero at all selected nonaxis points. Division by those multipliers transfers the obstruction to the full normalized observables. Evenness and conjugation supply every quadrant for LT. The two separately proved upper HT quadrants and conjugation supply every quadrant

for HT. Prime-grid density then rules out continuation at every point of the circle, including the four axis points: any proposed neighborhood of an axis point would also continue through a nearby obstructing nonaxis point. Finally $t = 1/s$ is biholomorphic at every unit-circle point and transfers the exterior HT assertion to its prescribed disk germ. □

7.7 The energy variable, the physical field, and covering

For $\lambda = 1$ the explicit profiles are

$$H_1(y) = \operatorname{arsinh}(\cos y), \quad |y| < \pi/2, \qquad L_1(y) = \operatorname{arsinh} |\cos y|.$$

The elementary identity

$$|\sinh(x + \mathrm{i}y)|^2 = \sinh^2 x + \sin^2 y \tag{7.56}$$

shows that $s = \sinh z$ maps Ω_H onto $|s| < 1$ and Ω_L onto $|s| > 1$ with the following exact mapping properties. On the disk, choose the analytic square root of $1 + s^2$ equal to one at zero. Then $\operatorname{arsinh} s = \log(s + \sqrt{1 + s^2})$, with value zero at zero, is analytic; its image lies in the component of $\{|\sinh z| < 1\}$ containing zero, which is exactly Ω_H. The inverse identities, first near zero and then by analytic continuation, prove that $\sinh : \Omega_H \to \{|s| < 1\}$ is a biholomorphism.

For LT set $w = e^z$ and $s = (w - w^{-1})/2$. For every $|s| > 1$, the two roots of $w^2 - 2sw - 1 = 0$ have product -1 and neither lies on $|w| = 1$; exactly one has modulus greater than one. This unique outer root depends holomorphically on s, since its only possible quadratic branch points would be $s = \pm\mathrm{i}$ on the excluded circle. Thus $w \mapsto (w - w^{-1})/2$ is biholomorphic from

$$\mathcal{W} = \{|w| > 1 : \ |(w - w^{-1})/2| > 1\}$$

onto $|s| > 1$. Identity (7.56) gives $\exp^{-1} \mathcal{W} = \Omega_L$. The epigraph Ω_L is simply connected, so the restriction of the exponential is its universal covering of $\mathcal{W}$. Consequently $\sinh : \Omega_L \to \{|s| > 1\}$ is the universal covering, with no omitted temperature sheets.

At every boundary point with $\cosh z \neq 0$, this change of variable is locally biholomorphic and takes the appropriate domain side to the appropriate side of the unit circle. Its only boundary ramification points have $z = \mathrm{i}(\pi/2 + \pi l)$; they are isolated on the LT graph and are the two HT tips. The prime obstructions lift densely on each remaining boundary arc. A neighborhood continuation at an isolated ramification point would therefore continue through a nearby lifted prime obstruction as well. The entire prescribed energy-variable boundaries follow, including these tips and all LT sheets.

Finally retain the physical field convention from Section 1: the Hamiltonian field has energy units, $z = 2\beta J_x$, and

$$\chi_H(z) = \frac{z}{2J_x} \mathcal{X}_H(z), \qquad \chi_L(z) = \frac{z}{2J_x} \mathcal{X}_L(z).$$

The thermodynamic argument identifies this response with the uniform energy-field derivative, with the plus phase fixed before the zero-field limit in LT. The connected sum in (7.3) includes the M^2 subtraction. The factor $z/(2J_x)$ is analytic and nonzero at every boundary point and preserves the local obstructions. In the LT exterior the local inverse $\operatorname{arsinh} s$ is read on the physical temperature cover.

Completion of the proof of Theorem 1.1. The unequal-ratio part was proved in Section 6. At equal coupling Theorem 7.11 proves the full normalized obstructions, and the covering and physical-field calculation above transfers them to the prescribed energy-variable domains. No limit in λ or phase duality is used. This proves the theorem for every finite positive ratio on both physical temperature sides. □

Corollary 7.12 (Inverse-temperature barriers)**.** *For each fixed $J_x > 0$, the barriers in the β plane are exactly $\partial\Omega_H/(2J_x)$ and $\partial\Omega_L/(2J_x)$ for their respective physical germs. At $\lambda = 1$ they are the correctly interpreted pullbacks of $|\sinh(2\beta J)| = 1$, with the low-temperature covering specified above.*

References

[1] A. I. Bugrij and O. Lisovyy, Correlation function of the two-dimensional Ising model on a finite lattice. II, *Theoretical and Mathematical Physics* **140** (2004), 987–1000. arXiv:0708.3643v1.

[2] N. Iorgov and O. Lisovyy, Ising correlations and elliptic determinants, *Journal of Statistical Physics* **143** (2011), 33–59. doi:10.1007/s10955-011-0154-6. arXiv:1012.2856v2.

[3] W. P. Orrick, B. G. Nickel, A. J. Guttmann and J. H. H. Perk, The susceptibility of the square lattice Ising model: New developments, *Journal of Statistical Physics* **102** (2001), 795–841. doi:10.1023/A:1004850919647. arXiv:cond-mat/0103074.

[4] S. Boukraa, S. Hassani and J.-M. Maillard, Holonomic functions of several complex variables and singularities of anisotropic Ising n-fold integrals, *Journal of Physics A: Mathematical and Theoretical* **45** (2012), 494010. doi:10.1088/1751-8113/45/49/494010. arXiv:1207.1784v2.

[5] A. Raoufi, Translation-invariant Gibbs states of Ising model: General setting, preprint (2017). arXiv:1710.07608v1.

[6] F. Camia, C. Garban and C. M. Newman, The Ising magnetization exponent on $\mathbb{Z}^2$ is 1/15, preprint (2013). arXiv:1205.6612v4.